MILITARY ENGINEER SERVICES

GARRISON ENGINEER [PROJECT]
STATION ROAD, VISAKHAPATNAM – 530 004

PROVISION OF 01 NUMBER BADMINTON COURT AT NAD SUNABED UNDER GE (P) NAVY VISAKHAPATNAM

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No. of drawings: NIL

SIGNATURE OF CONTRACTOR

AGE [CONTRACTS]
FOR ACCEPTING OFFICER

Tel: 0891 - 2546410
E-mail: gepvizag-ap@nic.in

Garrison Engineer [P] [N]
Military Engineer Services
Station Road
Visakhapatnam – 530 004

8891/ /E8 Aug 2025

Shri/ M/s _____

_____

_____

PROVISION OF 01 NUMBER BADMINTON COURT AT NAD SUNABED UNDER GE (P) NAVY VISAKHAPATNAM

Dear Sir(s),

1.0 Tender documents in respect of above work are uploaded on the site **www.defproc.gov.in.** The tender is on single stage two Cover e-tendering system. The contents of Cover – I & Cover – 2 are specified in Notice Inviting Tender.

2.0 Bids will be received online by Accepting Officer up to the bid submission end date and time mentioned online**.** No tender / bid will be received in physical form and any tender / bid received in such manner will be treated as non bonafide tender / bid.

3.0 Bid will be opened on due date and time fixed for opening in the presence of tenderers / bidders or their authorized representatives, who have uploaded their quotation / bid and who wish to be present at the time of opening the bids.

4.0 Your attention is also drawn to "Instruction on Filling and Submission of Tender" attached herewith. You may forward your points on tender documents and/or depute your technical representative for discussion on tender / drawings and to clarify doubts, if any, on or before bid submission start date. You are requested not to write piece meal points and forward your points duly consolidated before due date.

5.0 Un-enlisted contractor are required to submit the scanned copies [in pdf file] of documents required as per eligibility criteria mentioned in "Instruction on Filling and Submission of Tender" and Appendix "A" to NIT along with Earnest Money Deposit [EMD] & tender fee and submit the physical documents in the office of **Garrison Engineer [Project] [Navy], Station Road, Visakhapatnam – 530 004** within time limit specified in NIT. Inadequacy / deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6.0 Contractor having not executed Standing Security Bond and Standing Security Deposit in any MES formation shall upload scanned copy of **Earnest Money Deposit [EMD]** mentioned in Notice Inviting Tender and shall ensure receipt of hard copy of EMD in the office of tender issuing authority before dated & time fixed for this purpose. In case of failure to abide by any of these two requirements, the finance bid will not be opened.

7.0 Enlisted contractors of MES shall submit the scanned copies [in pdf file] of enlistment letter, tender fee and such other documents as mentioned in Appendix "A" to NIT and submit physical documents in the office of **Garrison Engineer [Project] [Navy], Station Road, Visakhapatnam – 530 004** before date & time fixed for this purpose.

8.0 The contractor must ensure that the tender/ bid on the proper form is uploaded in time as the Accepting Officer will take no cognizance of any quotations / offer received in any other electronic or physical form like e-mail / fax / by hand / through post from tenderer / bidder even if they are received in time.

9.0 In view of delays due to system failure or other communication related failures, it is suggested that the tender / bid be uploaded, if necessary, sufficiently in advance of the last due date and time fixed.

10.0 General Conditions of Contracts [IAFW – 2249] [1989 Print] and errata and amendments thereto, Schedule of Minimum Fair Wages [As per Government of India notification dated 10 March 1992], MES Standard Schedule of Rates, Part – I [2009] [Specifications] and MES Standard Schedule of Rates, Part – II [2020] [Rates] are not enclosed with these documents. These are available for perusal in the office of Garrison Engineer concerned and this office.

Note: Tenderers are deemed to be in possession of the above documents. The documents mentioned above are available in booklet form in any of the MES Offices, which can be seen during working hours on any working day, with prior notice. The tenderers are deemed to have made themselves acquainted with the contents of the above mentioned documents before submission of the tender / bid and no claim whatsoever on this account shall be entertained.

11.0 ANY TENDER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITION, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED**.**

12.0 Contractor shall get them registered with the Employees Provident Fund Organization and deposit the necessary contributions with the EPFO. Further, it is a mandatory requirement that all construction workers should be provided Universal Account Number [UAN] by the contractor by appropriately registering them on the EPFO portal. All the workers deployed by contractors are enrolled as members of Provident Fund and should be given the Universal Account Number [UAN]. Certificate to the effect that all the workers employed directly or indirectly are registered for EPF and the due consideration have been credited into their account shall be rendered by the contractor along with each and every RAR / Final Bill.

Note: For the purpose of above, contractor shall have Provident Fund Code Number obtained from concerned authorities [Employees Provident Fund Organization]. Incase tender/bid is accepted, the Provident Fund Code Number shall be submitted to GE before placing the Work Order.

13.0 **GST REGISTRATION NUMBER:** The contractor shall be in possession of GST Registration Number. It is mandatory for the contractors to upload their GST Registration Number along-with the Cover 1. Contractor, who does not upload GST Registration Number, shall be disqualified in the technical evaluation and his Finance Bid shall not be opened.

14.1 On acceptance of tender within 28 days of receipt of the Letter of Acceptance, the successful Contactor shall deliver to the Accepting Officer a Performance Security in the form of FDR / BGB for an amount equivalent to **5%** of the Contract Sum. (See amended Condition 19 of GCC (IAFW-2249). If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, the same shall be from Nationalized / Scheduled Indian Bank.

14.2 Failure of the successful Contractor to comply with the above requirement shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES Enlisted contractor, amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury. Not more than one tender/bid shall be submitted/uploaded by one contractor or one firm of contractors. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

Encls: (Tender Documents) Yours faithfully

SIGNATURE OF CONTRACTOR AGE [CONTRACTS]
FOR ACCEPTING OFFICER

INSTRUCTIONS FOR FILLING IN AND SUBMISSION OF E-TENDER DOCUMENTS TO BE COMPLIED WITH BY THE CONTRACTOR

1 **EARNEST MONEY DEPOSIT [EMD]:** Contractor[s] who are not enlisted with MES / who are enlisted but have not executed the Standing Security Bond shall submit "**Earnest Money Deposit"** as detailed in Notice Inviting Tender in one of the following forms, along with their tender/bid.

[a] Deposit at Call Receipt from a Scheduled Bank in favour of Garrison Engineer concerned.

[b] Receipted Treasury Challan, the amount being credited to the Revenue Deposit of Garrison Engineer concerned.

It is advisable that Earnest Money is deposited in the form of Deposit at Call Receipt from an approved Scheduled Bank for easy refund. In case the tenderer/bidder wants to lodge "Earnest Money Deposit" in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money shall be submitted in the name of concerned Garrison Engineer.

Note: Earnest Money Deposit [EMD] in the form of Cheque / Bank Guarantee etc. will not be accepted. Non-Submission of **Earnest Money Deposit [EMD]** [Scanned copy along with technical bid & hard copy before the date & time fixed for opening of Cover 1] WILL RENDER THE BID DISQUALIFIED FOR OPENING OF COVER – 2 [FINANCE BID].

2 **GENERAL INSTRUCTIONS FOR COMPLIANCE:**

2.1 The bids received only in the electronic form will be considered. All bids shall be submitted on **www.defproc.gov.in** portal. Documents should be scanned and forwarded in pdf form and xls from as indicated.

2.2 Bids shall be uploaded on www.defproc.gov.in portal on or before the bid closing date mentioned in the tender. No tender / bid in any other electronic or physical form like e-mail / fax / by hand / through post will be considered.

2.3 Bid should be DIGITALLY signed using valid DSC. All pages of tender documents, corrections/alterations shall be signed / initialed by the lowest bidder after acceptance.

2.4 Drawings, if issued in physical form, must be returned duly initialed by the tenderer / bidder in separate envelope indicating his name and address.

2.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialed. The contractor shall initial every page of tender and sign all drawings forming part of the tender. Any tender / bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

INSTRUCTIONS FOR FILLING IN AND SUBMISSION OF E-TENDER [CONTD…]

2.6 In the technical bid, a scanned copy of Power Attorney in favour of the person uploading the bid using his / her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect, stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in pdf form. In case of partnership concern or a limited company, digital signatory of the bid / tender shall ensure that he is competent to bind the contractor [through partnership deed general power of attorney or Memorandum and Articles of Association of the Company] in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of such authority shall be attached with the tender / bid in pdf form, if not submitted earlier. The person uploading the bid on behalf of another partner[s] or on behalf of a firm or company using his DSC shall upload with the tender / bid a scanned copy [in pdf form] of Power of Attorney duly executed in his favour by such other or all of the Partner[s] or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company as the case may be, in all matters pertaining to the contract including the Arbitration Clause.

2.7 Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the Firm or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in pdf form with the tender/bid unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the Constitution or the Company as laid down in its Memorandum & Article of Association.

2.8 Hard copies of all above documents should be sent by the Tender issuing authority well in advance to be received before the date & Time fixed for the same.

2.9 Bid [Cover – 1 & Cover – 2] shall be uploaded online well in time.

2.10 The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to Special Condition – 3 referred hereinafter and also Conditions 24 & 25 of IAFW – 2249 [General Conditions of Contracts].

2.11 Tenderers / bidders who uploaded their priced tenders / bids and are desirous of being present at the time of opening of the tenders / bids, may do so at the appointed time.

2.12 The tenderer / bidder shall quote his rate on the BOQ file only. No alteration to the format will be accepted; else the bid will be disqualified and summarily rejected.

2.13 In case the tenderer / bidder has to revise / modify the rates quoted in the BOQ [Excel Sheet] he can do so only in the BOQ, through www.defproc.gov.in site only before the bid closing time and date.

3.0 **REVOCATION / REVISION OF OFFER UPWARD / OFFERING VOLUNTARY REDUCTION, AFTER OPENING OF FINANCIAL BIDS BY LOWEST BIDDER:** In the event of lowest tenderer / bidder revoking his offer or revising his rates upward / offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and **Earnest Money deposited** by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer / bidder for depositing the amount through MRO. Bids of such Contractors / bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Government Treasury. In addition, bids of such tenderer / bidder and his related firm shall not be opened in second call or subsequently calls. Reduction offered by the tenderer / bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

INSTRUCTIONS FOR FILLING IN AND SUBMISSION OF E-TENDER [CONTD…]

4.0 Department may issue amendments / errata in from of CORRIGENDUM to tender / revised BOQ to the tender documents. The tenderer / bidder are requested to read the tender documents in conjunction with all the errata / amendments / corrigendum, if any issued by the department.

5.0 **PERFORMANCE SECURITY DEPOSIT**:

5.1 Within 28 days of receipt of the letter of Acceptance, the successful contract shall deliver to the Accepting Officer a **Performance Security Deposit** in any of the forms given below **for an amount equivalent to 5% of the contract sum**.

[a] A **Bank Guarantee** in the prescribed form.

[b] Government Securities, **FDR** or any other Government Instruments stipulated by the Accepting Officer.

5.1.1 The full amount of Performance Security Deposit for each contract shall be given by the contractor in any one of the forms stipulated above, mix of forms for the same contract shall not be allowed.

5.1.2 The period of validity of the Bank Guarantee Bond / FDR against Performance Security Deposit shall be upto and including the scheduled date of expiry of Defects Liability Period.

5.2 If the performance security is provided by the successful contractor is in the form of a Bank Guarantee. It shall be issued by Nationalized / Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

5.3 Failure of the successful Contractor to comply with the above requirement shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES Enlisted contractor, amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury. Not more than one tender/bid shall be submitted/uploaded by one contractor or one firm of contractors. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5.4 All compensation or other sums of money payable by the contractor to the Government under the terms of this contract or under any other contract with Government may be deducted from, or paid by sale of a sufficient part of the Performance Security or from the interest arising there from or from any sums which may be due or become due to the contractor by the Government on any account whatsoever and in the event of his Performance Security being reduced by any reason of any such deduction or sale as aforesaid, the contractor shall within ten days thereafter make good in cash or securities, endorsed as aforesaid, any sum or sums which may have been deducted from or realized by the sale of his Performance Security or any part thereof.

5.4.1 Government shall not be responsible for any loss of securities or any depreciation in the value of securities while in their charge nor for loss of interest thereon.

5.5 In the event of contract being cancelled, under Condition 52, 53 and 54 of General Conditions of contract IAFW-2249, **the Performance Security and Retention Money** upto last paid RAR shall be forfeited in full and shall be credited into Consolidated Fund of India. All T&P and material of contractor lying at site shall be confiscated by the Government and shall be absolutely at the disposal of the President of India and No compensation whatsoever shall be allowed by the department and the firm shall be barred from participating in the tender for the balance work.

INSTRUCTIONS FOR FILLING IN AND SUBMISSION OF E-TENDER [CONTD…]

6.0 **DEFECTS LIABILITY PERIOD:** The defect liability period for subject work is **24 Calendar Months** after Certified date of Completion of the work as per amendment No. 49 issued to condition 46 of IAFW 2249 General Conditions of Contracts.

7.0 **REFUND OF PERFORMANCE SECURITY:** The Performance Security Deposit mentioned above may be refunded to the contractor after expiration of the defects liability period Twenty four calendar months after the works have been handed over to Government] by the GE provided always that the contractor shall first have been paid the final bill and have rendered a No Demand Certificate [IAFW-451]. If any recovery is outstanding against the contractor, the release of Performance Security Deposit is subject to the compliance of the procedure for effecting recovery / withholding the due amount as stipulated in Condition 67 of IAFW-2249 as amended.

8. Tenderers attention is drawn to the official secret Act, particularly section 5 thereof for compliance.

9. These instructions shall form part of the contract documents and shall be signed along with the tender documents during signing of contract agreement.

SIGNATURE OF CONTRACTOR

AGE [CONTRACTS]
FOR ACCEPTING OFFICER

[In lieu of IAFW – 2162] [Revised – 1960]

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER

1.0 A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT).

2.0 The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate however, is not a guarantee and is merely given as a rough guide and if the work cost more or less, a tenderer/bidder will have no claim on that account.

3.0 The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handling over site, which will be on or about two weeks after the date of Acceptance of tender.

4.0 Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and unenlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. However in case of term contracts, enlisted contractors of Class SS to E may submit tender. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5.0 The Officer of Garrison Engineer (Project) (Navy) Vizag will be the Accepting Officer here in after referred to as such for purpose of the contract.

6.0 The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the contractor on or before the date & time mentioned in **NIT**. A scanned copy of DD/BC with enlistment details and other documents as specified in Appendix A shall be uploaded as Cover-1 (Technical bid) of the tender on e-tendering portal. DD/BC is refundable in case the contractor is not considered eligible in technical evaluation of Cover-1 resulting in non-opening of Cover-2. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD/BC including revalidation of DD/BC and shall not have any claim from Government whatsoever on this account.

7.0 Tender form and conditions of contract and other necessary documents shall be available on website "defproc.gov.in" for downloading and shall form part of contract agreement in case the tender/bid is accepted.

8.0 In case of MES enlisted contractor who has not executed the Standing Security Bond and un enlisted contractor, the Cover-I shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CCE/GE/GE (I)/AGE (I) (see Appendix 'A') by a Scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of the concerned CCE/GE/GE(I)/AGE(I) (see Appendix 'A').

9.0 The CCE/GE/GE (I)/AGE (I) will return the Earnest Money, wherever applicable, to all unsuccessful tenderers/bidders by endorsing an authority on the deposit at call receipt for it's refund, on receipt of intimation from the Accepting Officer to do that.

10.0 In case of successful contractor i e the lowest contractor having submitted EMD, he shall have the option of converting the EMD instrument into part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tender from Accepting Officer.

NOTICE INVITING TENDER [Contd….]

11.0 Copies of the drawings and other documents pertaining to the work signed for the purpose of identification by the Accepting Officer or his accredited representative, sample of materials and stores to be supplied by the contractor will also be available for inspection by the tenderer/bidder at the office of Accepting Officer and concerned GE/ GE(I)/ AGE(I) during working hours.

12.0 The tenderers / bidders are advised to visit the site of work by making prior appointment with GE/ GE(I)/ AGE(I)/ CCE/ Project Manager who is also the Executing Agency of the work (see appendix 'A'). The tenderers/bidders are deemed to have full knowledge of all relevant documents, samples, site etc., whether they have inspected them or not.

13.0 Any tender/bid which proposes any alteration to any of the conditions laid down or which proposes any other condition or prescription Whatsoever, is liable to be rejected.

14.0 The uploading of bid by a bidder implies that bidder has read this notice and the Conditions of Contract and has made himself aware of the scope and specifications of work to be done and of the conditions and rates at which stores, tools and plants etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

15.0 The bidder must be in possession of a copy of the MES Schedule (SSR) (Part-I & Part- II of latest edition) including amendments and errata thereto.

16.0 Invitation for e-tender does not constitute any guarantee for validation of 'T' bid and subsequent opening of finance bid of any applicant/bidder, even of enlisted contractors of appropriate class, merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the 'T' bid and not open the finance bid of any applicant/bidder. 'T' bid validation shall be decided by the Accepting Officer based on, interalia and capability of the firm as per criteria given in Appx 'A' to this NIT. The applicant contractor/bidder will be informed regarding non-validation of his 'T' bid assigning reasons thereof through the "defproc.gov.in" website. The applicant contractor/bidder if he so desires may appeal to the next higher Engineer authority i.e. HQ, Commander Works Engineers No.2, Station Road, Visakhapatnam – 530004 with copy to the Accepting Officer on email before the scheduled date of opening of Finance Bid. The decision of the Next Higher Engineer Authority (NHEA) shall be final and binding. The contractor/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

17.0 The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking, giving a price preference over other Tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible from such tenderer/bidder whose tender/bid is rejected.

18.0 Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

19.0 This **Notice Inviting Tender (NIT)** including Appendix 'A' and Annexure's thereto if any shall form part of the contract agreement.

SIGNATURE OF CONTRACTOR

GARRISON ENGINEER [P] NAVY
FOR ACCEPTING OFFICER

APPENDIX 'A' TO NOTICE INVITING TENDER [NIT]
(FOR WORKS COSTING LESS THAN RS. 15 CRORES)

1.	Name of work	:	**PROVISION OF 01 NUMBER BADMINTON COURT AT NAD SUNABED UNDER GE (P) NAVY VISAKHAPATNAM**
2.	Estimated cost of work	:	**Rs. 43.30** Lakhs [At par Market]
3.	Period of Completion	:	**06 (Six) Months**
4.	Cost of tender documents & Garrison Engineer	:	Rs. **500/-** in the shape of Demand Draft (DD)/ Bankers Cheque from any Nationalized/ Scheduled Bank in favour of Garrison Engineer [Project] [Navy] Visakhapatnam Note: In case of retendering the contractor, who had quoted in the previous call is not required to submit the cost of tender
5.	Website / portal address	:	www.defproc.gov.in and www.mes.gov.in
6.	Type of contract	:	The tender shall be based on IAFW – 2249 and IAFW – 1779-A [Revised-1955] with Schedule "A" [Item of works] to be priced by the tender
7.	Information & Details: [a] Bid submission start date [b] Last date of bid submission [c] Date of bid opening		Refer critical dates
8.	Eligibility criteria:		
	[A] For MES enlisted contractors	:	All contractors enlisted with MES in Class **'D'** and Category **'a (i)'** subject to satisfactory remarks w.r.t performance in respect of works in hand reflected in Work Load Return [WLR] or any other report circulated by competent engineer authority.
	[B] For other contractors	:	[a] The firms not enlisted with MES shall meet the enlistment criteria of **'D'** class MES contractors & **a (i)** category contractor with regard to satisfactorily completion of requisite value of works with Central/State Government/ Central/ State PSUs/AWHO/AFNHB/CGEWHO/DGMAP, Annual turnover, Bank Solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in), [b] Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. [c] Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date

APPENDIX 'A' TO NOTICE INVITING TENDER [CONTD....]

9.	Tender issuing and Accepting Officer	:	Garrison Engineer [Project] [Navy], Station Road, Visakhapatnam. Email: **gepvizag-ap@nic.in** **Tele: 0891 2546410**
10.	Executing Agency	:	AGE [B/R] Sunabeda,Odisha.
11.	Earnest Money	:	**Rs. 86,600/-** in favour of Garrison Engineer [Project] [Navy], Visakhapatnam.

NOTES:-

[1] In case after opening cover 1, if the number of MES enlisted contractors of eligible class as well as eligible un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT are less than 07 (Seven), applications in respect of MES contractors of one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE command / ADG) below the eligible class shall also be considered subject to fulfillment of other eligibility criteria given in the NIT. Therefore MES contractors one class below (two classes below in case of remote and difficult areas) may also bid for this tender. Such contractors (contractors of one/two classes below the eligible class) shall not be considered in case their present residual work in hand is more than FIVE TIMES their present tendering limit. However in case such contractors fulfil the criteria of upgradation to the stipulated eligible class based on past experience of completed works (individual work experience and/or average annual turnover, as applicable) and financial soundness (solvency/financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-1 bid details related to residual work in hand like details of work in hand showing names of work, names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfil the criteria of upgradation shall also upload the requisite information/documents in support of upgradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

[2] In case after opening of Cover-1, the number of MES enlisted contractors of eligible class as well as unenlisted contractors, if any fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed
Similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of completed works (individual work experience and/or average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore such contractors shall upload the requisite information/documents in the Cover-1

[3] Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national/ Indian Nationals staying abroad/ India National having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

[4] Contractors enlisted with MES will upload following documents in Cover-1 for checking eligibility:-

[a] Application for tender on Firm's letterhead.

[b] Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.

[c] Scanned copy of DD/ Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.

[d] Any other document required as described in this Appendix.

[5] Contractors not enlisted with MES will be required to upload following documents in Cover-I for checking eligibility:-

[a] Application for tender on Firm's letterhead.

[b] Scanned copy of DD/ Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.

[c] Copy of Police Verification Report/ Police Clearance Certificate/ Character Certificate from Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of proprietor/each Partner/Each Director.

[d] All documents required for enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.5 of Section 1 of MES Manual on Contracts 2020

[e] Details of works being executed in MES, if any

[f] Any other document required as described in this Appendix

[6] Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover-1 shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.

[7] Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within 07 (Seven) days of bid submission end date failing which following action shall be taken:-

[a] In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid (Cover 2).

[b] In case of tenders from un-enlisted contractor, where scanned copies of requisite DD/ Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial Bid (Cover 2).

[c] In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tender shall not qualify for opening of financial bid (Cover 2).

[8] Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/Director and firms own employees, director, project manager **provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.**-

APPENDIX 'A' TO NOTICE INVITING TENDER [CONTD….]

[9] After opening of Cover 1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of email/ SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from the date of communication, failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same

[10] Invitation for e-tender docs not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer authority (NHEA) viz COMMANDER WORKS ENGINEER NO.2 on email id **cwepvskp-ap@nic.in** with copy to the Accepting Officer on email before the scheduled dated of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his

[11] In case an un-enlisted contractor is already executing works in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES class of contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in the Cover 1 of the bid and shall be checked/verified by the Accepting Officer.

[12] In case the BOQ is revised through the corrigendum and bidder has failed to quote on revised BOQ (ie he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such case the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

[13] Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a wilful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an un-enlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond an amount equal to the Earnest Money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In case a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if an, shall not be eligible for this tender in second call or subsequent calls.

APPENDIX 'A' TO NOTICE INVITING TENDER [CONTD....]

[14] Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

[15] The successful tenderer [either enlisted or un-enlisted] is required to lodge Performance Security for an amount equal to **5%** of the Contract Sum within 28 Days of receipt of Letter of Acceptance in the form of Bank Guarantee in the prescribed format or Government Securities or FDR or any other Government Instruments with the Accepting Officer in terms of Condition – 19 of IAFW – 2249, General Conditions of Contracts. The Performance Security shall be valid till expiry of Defects Liability Period as stipulated here-in-after PLUS Two Months. In case the tenderer fails to deposit the Performance Security within the stipulated period, the award of work shall be cancelled and Earnest Money deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderer shall remain suspended till the aforesaid amount equal to Earnest Money is deposited in Government Treasury. The contractor shall not be entitled for any compensation whatsoever on this account. The Work Order for commencing the work shall be placed by the GE only after receipt of above Performance Security by the Accepting Officer within the stipulated time period.

SIGNATURE OF CONTRACTOR

GARRISON ENGINEER [P] NAVY
FOR ACCEPTING OFFICER

(In lieu of IAFW-1779A) (Revised – 1955)

(TO BE USED IN CONJUNCTION WITH GENERAL CONDITIONS OF CONTRACT IAFW-2249 (1989 PRINT)

MILITARY ENGINEER SERVICES

Tel: **0891 - 2546410**
E-mail: gepvizag-ap@nic.in

Garrison Engineer [P] [N]
Military Engineers Services
Station Road
Visakhapatnam – 530 004

8891 / /E8 Aug 2024

TENDER AND CONTRACT FOR WORKS BY MEASUREMENT

PROVISION OF 01 NUMBER BADMINTON COURT AT NAD SUNABED UNDER GE (P) NAVY VISAKHAPATNAM

M/s _____ is here by authorized to tender for the above work.

The complete tender is uploaded on MES portal of www.eprocuremes.gov.in. The bidders shall submit their bids through **www.eprocuremes.gov.in.** The bid will be opened in office of Garrison Engineer [Project] [Navy], Station Road, Visakhapatnam - 530 004. The financial bids of only those tenderers who meet the eligibility criteria as per Notice of tender shall only be opened.

Any correspondence concerning this tender should be addressed as indicated at the top of this sheet quoting the reference as given.

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER

(SIGNATURE OF CONTRACTOR)

(SIGNATURE OF THE OFFICER ISSUING THE TENDER DOCUMENTS)

Appointment: **AGE [CONTRACTS]**

For Accepting officer

SCHEDULE 'A' NOTES

1A. The repairs covered under scope of work is as under:-

PROVISION OF 01 NUMBER BADMINTON COURT AT NAD SUNABED UNDER GE (P) NAVY VISAKHAPATNAM

1. The entire work covered under this contract i.e. Schedule 'A' shall be completed within a period of **06 (Six)** months from the date of handing over of site.

2. The description of items of works in BOQ are in brief. These shall be read together with Drawings, Specifications, Special conditions, notes, method of measurements, preambles, etc., as given in MES Schedule and General Conditions of Contracts (IAFW-2249) shall be followed except where specifically provided for otherwise. The intention of Accepting Officer in this regard shall be final, conclusive and binding on the Contractor.

3. Quantities inserted in the BOQ are provisional and are given as guidance only for tenderers. These shall, however not be varied beyond the limits laid down in Condition 7 of General Conditions of Contracts (IAFW-2249).

4. Tenderer shall insert his unit rates in figures only under relevant column in BOQ. The rates inserted shall be for material and labour/supply & fix complete unless otherwise specifically mentioned in the description of respective items.

5. The total amount in the BOQ is not firm but will be treated as contract sum as referred to in IAFW-2249, General conditions of contracts.

6. The rates quoted by the tenderer in BOQ shall be deemed to include for any minor details of works and/or construction which are obviously and fairly intended and which may not have been specifically referred to in the tender documents but which are essential for the execution and completion of the work in a workman like manner and sound construction. In case of difference of opinion between the contractor and Garrison Engineer as to whether it constitutes minor or extra constructional detail included in the unit rate quoted or not, the decision of the Accepting Officer, shall be final, conclusive and binding.

7. Tenderers to note that unit rates / amount quoted in the BOQ shall deemed to be inclusive of all taxes/cesses viz GST, duties, royalties, octroi and other levies payable under respective statutes as applicable on the date of Bid submission end date. Nothing extra shall be paid to the contractors on account of any misunderstanding regarding applicability of taxes. It may be noted that any tender imposing any condition in this regard or on any other account shall be treated as a conditional tender and the same shall be liable to be rejected.

8. The contractor should keep in mind while quoting their/his tender, **the fact that the buildings / site where the repair/work is to be carried out are under occupation and shall remain under occupation while carrying out the work**. The contractor will carry out the work in phased manner in consultation with Engineer-in-Charge. It will be the responsibilities of the contractor to submit in writing his planning program of work seven days in advance to take over site/commence the work or any other requirements. No compensation or extra shall be admissible of any misunderstanding or loss of man-hours.

9. The works shall be executed by **Garrison Engineer [Project] [Navy] Visakhapatnam**. The word GE or Garrison Engineer wherever occurs in the tender documents shall be read as **Garrison Engineer [Project] [Navy] Visakhapatnam**.

10. License for carrying out electrical works: It shall be made mandatory that contractors who apply for work should have a valid license holder in their establishment and his license should be uploaded in cover-1 of the bid or otherwise financial bid of tender shall not be opened.

11. The subject work covered under Residential and Depot area of NAD Sunabeda. The scope of work covered under the subject work is regular repair/replacement of Int Ele & street lights by attending day to day complaints. The quoted rates in the BOQ shall be inclusive of the above provisions.

12. (a) The quoted Unit rate shall be deemed to be inclusive of all type of taxes/cesses viz GST, state sales taxes, central sales taxes, actual surcharge, labour welfare tax, state octori tax & Swachh Bharat Tax and any other type of prevailing levies including GST on materials etc.

(b) **The quoted Unit rate shall be deemed to be inclusive of all taxes/cesses viz GST implemented by Central Govt. w.e.f. 1st July 2017, duties, royalties, octroi and other levies payable under respective statutes as applicable on the date of receipt of tender. It may be noted that any tender imposing any condition in this regard or on any other account shall be treated as a conditional tender and the same shall be liable to be rejected. Any abolition of previous tax due to implementation of GST by Govt of India shall also be accounted while quoting the rate and accordingly rates quoted are deemed to be included this aspect. The contractors are advised to take this fact in mind while quoting tender. Nothing extra whatsoever on this account is admissible.**

(c) The rate quoted by contractor shall also includes all taxes/cesses viz GST/duties i.e trade tax on works contracts, excise duty, custom duty, octroi duty & any other taxes/duties as applicable. From 3D, 31 & 32 or any other form shall not be issued by MES. This is an express condition of this no claim of the contractor shall be entertained by MES on this account.

(d) It is incumbent on the part of tenderers to get them self registered with Central Board of Excise and custo,s. Irrespective of whether any tenderer/ Bidders has got his firm registered or not for execution of the work contract, the GST shall be deducted at source the sum equivalent to percentage of the payment made to the contractor as per Govt Policy

13. **GOODS & SERVICE TAX (GST)** : For assets created under this contract, Goods & Service Tax (GST) shall be applicable as per Govt of India notifications. The tenderer should keep this in mind while quoting their offer.

14. **PERFORMANCE SECURITY.** Within 28 days of receipt of letter of Acceptance, the successful unenlisted/enlisted contractor would be required to deposit performance security in any of the forms given in clause 19 of IAFW-2249 for an amount equal to 5% of the contract sum. Failure of the successful contractor to comply with the requirements of sub-clause 19.1 of IAFW-2249 shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor, amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury**.**

15. **DEFECT LIABILITY PERIOD.** The defect liability period of the work shall be **two years (24 Months)** from the certified date of completion as per condition 46 of IAFW-2249.

SIGNATURE OF CONTRACTOR

AGE [CONTRACTS]
FOR ACCEPTING OFFICER

SCHDULE 'A'

BOQ

18 To 24

SCHEDULE 'B'

LIST OF MATERIALS TO BE ISSUED TO THE CONTRACTOR
(REFER CONDITION 10 OF 1AFW-2249)

Ser No.	Particulars	Unit	Rate at which materials etc. will be issued to Contractor (Rate in Rupees).	Place of issue by name	Remarks

-NIL-

SCHEDULE 'C'

ISSUE OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH WILL HIRED BY THE CONTRACTOR
(REFER CONDITION 15, 34 AND 35 OF IAFW-2249)

Sr. No.	Quantity	Particulars	Details of MES Crew supplied	Hire charges per unit per working day	Stand by charges per unit per day	Place of Issue	Remarks

-NIL-

SCHEDULE 'D'

TRANSPORT TO BE HIRED BY THE CONTRACTOR
(REFER CONDITION 16, AND 35 OF IAFW-2249)

Sr. No	Quantity	Particulars	Rate per unit per working day	Stand by charges per unit per day	Place of Issue	Remarks

-NIL-

SIGNATURE OF CONTRACTOR

AGE [CONTRACTS]
FOR ACCEPTING OFFICER

TENDER

THE PRESIDENT OF INDIA

Having examined and persuade the following documents:-

1. Specification signed by the Accepting Officer.

2. Drawing detailed in the specifications.

3. Schedule 'A', 'B', 'C' and 'D' attached hereto.

4. MES Standard Schedule of Rates 2009 [Part – I] "Specifications" together with Amendment No. 1 to 3 and MES Standard Schedule of Rates 2020 [Part – II] including amendment Nos. 1 to 122 "Rates" here-in-after referred as the MES Schedule together with amendments no. 1 to 122 for Part-II as applicable to the above schedule.

5. General conditions of contracts IAFW - 2249 (1989 Print) together with and including amendment Nos. **1 to 49** and Errata **1 to 20.**

6. **WATER CONDITION 31 OF IAFW-2249:GENERAL CONDITIONS OF CONTRACT**

Water will be supplied by the MES and shall be paid for by the Contractor @ Rs. 3.75 per 1000 worth work done priced at contract rate.

7. Should this tender be accepted I/We agree:-

** (a) That the sum of **Rs 96,740.00 (Rupees Ninety Six thousand Seven hundred Forty only)** forwarded as earnest money shall be either be retained as a part of Security Deposit or refunded by the Government and receipt of the appropriate amount of Security Deposit all as per condition 22 of IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and conditions contained or referred to therein at the item rates contained in the afore said Schedule 'A.' OR, at such other rates as may be fixed under the provisions of condition 62 of IAFW-2249 and to carry out such deviations as may be ordered vide condition 7 of IAFW-2249 upto a maximum of **10% (Ten)** percent and further agree to refer all disputes, as required by the condition 70 and 71 of IAFW-2249 to the sole arbitration of a serving officer not below the rank of Dir/Col/Dir (Contracts) having done at least one executive tenure of minimum three years or having at least two years' experience of handling contract matters from the Panel Arbitrators approved by CE Command, whose decision shall be final, conclusive and binding.

** to be deleted where not applicable.

T E N D E R (CONTD……)

Signature _____Name _____

in the capacity of _____duly authorized to sign the tender for and on behalf of

M/S_____

(IN BLOCK LETTERS)

Date:

Postal Address_____
_____
_____

Telegraphic address_____

Telephone No._____

WITNESS ADDRESS

ACCEPTANCE

_____ Alterations have been made in the documents as evidence that these alterations were made before the execution of the contract agreement; they have been initialed by the **Contractor** and **Shri S Bhava Harana, AGE [Contracts].**

The said officer is hereby authorized to sign and initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on behalf of the President of India at the Item rates contained therein for the sum of Rs._____
(Rupees_____
_____only) on
_____ day of _____2025.

Signature: _____

(SHASHWAT SHUKLA)
MAJ
GARRISON ENGINEER [P]

Dated: this _____day of _____ 2025

Appointment:
Garrison Engineer [Project]
Station Road, Vishakhapatnam

(FOR AND ON BEHALF OF THE PRESIDENT OF INDIA)

GENERAL CONDITIONS OF CONTRACTS AND IAFW-2249 (1989 PRINT)

A copy of GENERAL CONDITIONS OF CONTRACTS (IAFW-2249) (1989 Print) with Errata 1 to 20 and amendments Nos. **1 to 49** including latest amendments issued has been supplied to me / us and is in my/our possession. I/we have read and understood the provision contained in the aforesaid GENERAL CONDITION OF CONTRACTS before submission of this tender and I/We * shall aside by the terms and conditions thereof as modified , if any elsewhere in these tender documents.

It is hereby further agreed and declared by me/us* that the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249-1989 PRINT) including condition 70 thereof pertaining to settlement of disputes by Arbitration, containing 33 pages with amendments Nos. **1 to 49** including latest amendments issued and errata 1 to 20 form part of these tender documents.

- Delete whichever is NOT APPLICABLE

AGE [Contracts]
For Accepting Officer

Signature of Contractor

SCHEDULE OF MINIMUM FAIR WAGES

1. It is hereby agreed that the Schedule of Minimum fair wages (SMFW) as published vide Government of India Notification dated 14 Oct 1988 and as amended subsequently from time to time by the orders of the competent Central / State Government / local authorities (comprising 14 pages) form part of the tender documents .

2. Schedule of minimum fair wages is not enclosed along with tender but the contractor is deemed to have full knowledge regarding the minimum wages payable to laborers as legally effective on date of submission of tender and his tendered rates shall be deemed to have been based on this irrespective of the fact that Gazette Notification may not have been published or that he was unaware of the increase in minimum wages. The minimum wages legally effective on the date of receipt of tender shall be the basis.

3. The minimum wages legally effective referred to above be the minimum wages notified in Gazette/Governed by any local regulations or by Central Government whichever is higher.

4. My / our signature hereunder amount to my/our having signed the aforesaid documents forming part of this tender.

NOTE: 'Schedule of minimum fair wages 'referred to above is available for reference in the office of The Garrison Engineer [P] [N], Station Road, Visakhapatnam - 4.

AGE [CONTRACTS]
For Accepting Officer

Signature of Contractor

SPECIAL CONDITIONS

1.0 GENERAL

1.1 The following conditions shall be read in conjunction with the General Conditions of contracts. IAFW –2249 (1989 Print) and IAFW –1779-A (Revised 1955) including Errata /Amendments there to. If any provision in these special conditions are at variance with that of the aforesaid documents the former shall be deemed to take precedence there over.

1.2 The work under this contract shall be carried out in accordance with Schedule 'A', the particular specifications , drawings and other provisions in MES Schedule.

1.3 The term "General Specifications" Special Conditions and all preambles in the MES Schedule shall be deemed to apply to the works under this contract. In case of any variance the provisions in these documents shall take precedence over the aforesaid provisions in the MES Schedule.

2.0 ADMISSION TO SITE BY CONTRACTOR TO ASCERTAIN HIS OWN INFORMATION

2.1 The tenderers shall contact the concern GE for the purpose of inspection of site(s) and relevant documents other than those sent herewith, who will give reasonable facilities for this purpose the tenderers shall also make themselves familiar with working conditions, accessibility of site(s), availability of materials and other cogent condition etc., which may effect the entire completion of work under this contract.

2.2 The tenderer shall be deemed to have visited the site(s) and made themselves familiar with the working conditions accessibility of site(s), availability of materials and other cogent condition etc. , which may effect the entire completion of work under this contract, whether he actually inspect the site(s) or not. No extra payment consequent on any mistake or misunderstanding or otherwise on this account will be allowed.

2.3 Contractor shall make suitable approaches to work site if required to convey men, materials plants and machinery required for the satisfactory completion of work. On completion of work the site shall be cleared off all approaches. The tendered rates shall interalia be deemed to include for the aforesaid contingencies.

3.0 SECURITY AND PASSES

3.1 Contractor's attention is invited to condition 25 of IAFW –2249. He shall employ only Indian Nationals, after verifying their antecedents and loyalty. The contractor shall on demand by the Engineer –in –Charge submit list of his agents employees and working people concerned and shall satisfy the Engineer –in –Charge as to the bonafide of such people.

3.2 The Engineer-in-Charge shall at his discretion have the right to issue passes as per rules and regulation of the installation area in force to control the admission of the contractor, his agents, and employees and work people to the site of the work or any part thereof. Passes should be returned at any time on demand by the Engineer –in –Charge or the authorities concerned and in any case on completion of work.

3.3 The contractor and his agents, employees and work people shall observe all the rules promulgated by the authority controlling the installations/area in which the work is to be carried out e. g. prohibition of smoking and lighting, fire precautions, search of persons on entry and exit, keeping to specific routes, observing specific timing etc., Nothing extra shall be admissible for any man hours etc., lost on this account.

4.0 CONDITIONS OF WORKING IN RESTRICTED AREA

4.1 The work lies in restricted area.

4.2 **VISIT OF SITE IN THE RESTRICTED AREA**

4.2.1 Permission to enter the restricted area at the time of submission of tenders can be obtained through the concern GE. Tenderers are advised to send prior intimation of their agents representatives etc., if any dates and time of their proposed visit so that necessary arrangements may be made by concern GE to ensure admission Whether tenderer visits the site or not, he shall be deemed to have full knowledge of the restriction of entering into exit from and working within the restricted area.

4.3 **ENTRY /EXIT**

4.3.1 The contractors, his agents, representatives workman etc., and his materials, carts, trucks or other means of transport, etc., will be allowed to enter through and leave from only such gates and at such times as the concern GE or authorities in charge of the restricted areas may at their sole discretion permit to be used. The contractor's authorized representative is required to be present at the place of entry and exist for the purpose of identifying his carts, trucks etc., to the personnel in charge of the security of restricted area.

4.4 **PASSESS OR IDENTITY CARDS**

4.4.1 The contractor, his agents and representatives are required individually to be in possession of an identity card or pass issued by the authorities through concern GE regarding workmen the contractor shall submit to GE the list of workmen and other details for issuing passes through the unit controlling authorities. These passes will be examined by the Security staff at the time of entry into or exit from the restricted area and also any time or number of times inside the restricted area, the tenderer shall have no claim on this account from the department.

4.4.2 Contractor shall be responsible for the conduct and action of his workmen, agents or representatives.

4.5 **SEARCH**

4.5.1 Thorough search of all persons and transport shall be carried out at each gate and for as many times as a gate is used for entry or exit and may also be carried out and any time or any number of times of the work site within the Restricted area.

4.6 **FEMALE SEARCHER**

4.6.1 If the contractor desires to employ female labour on works to be carried out inside the area of factory, depot, park etc. , and female searcher is not borne on the authorized strength of the Factory , Depot, Park etc. , at the time of submission of the tender, he shall be deemed to have allowed in his tender for pay and allowances etc. , for a female searcher (Class IV/Group 'D' Servant) calculated for the period female labour is employed by him inside that area. If more than one contractor employs female labour in addition to the authorized strength of the Factory, Depot, Park, etc. , the salary and allowances paid to the additional female searcher shall be distributed on an equitable basis between the contractors employing female labour taking into consideration the value and period of completion of their contracts. The GE decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

4.7 **WORKING HOURS**

4.7.1 The units controlling restricted area, usually, work during six days in the week and remain closed on the 7th day. The working hours available to contractor's labour and staff are however, appreciably reduced because of the time taken in security checks observed at the time of entry, exit and during working hours.

4.7.2 The exact working hours, days and non working days observed for the restricted area, where works are to be carried out shall be deemed to have been ascertained by the contractor before submitting his tender. The tenderers attention is invited to the fact that the total number of working hours for a unit are prescribed in regulations and they cannot be increased by the Garrison Engineer.

4.7.3 Contractor's materials, transport etc shall normally be permitted to come in/ go out of the area, between 9 AM to 5PM only and this time also may be reduced by the concerned unit authorities due to security reasons and contractor shall not have any claim on this account.

4.7.4 Contractor may also be allowed to carry out the work beyond 6 PM and up to 6 AM (day and night) with prior approval of GE. No claim of contractor whatsoever shall be entertained if such permission is not given by GE due to security reasons of the area. No movements of materials and transport to/out of site of works shall be permitted during night, unless special permission is obtained from the factory/ unit authorities.

4.8 **WORKS ON HOLIDAYS :** The contractor shall not carry out any work on gazette holiday, weekly holidays and other non working days except when he is specially authorized in writing to do so by the GE. The GE may at his sole discretion declare any day as holiday or non working day without assigning any reason for such declaration.

4.9 **ACCESS TO RESTRICTED AREA AFTER COMPLETION OR WORKS.**

4.9.1. After the works are completed and surplus stores etc removed, the contractor, the contractor, his agents, representatives, workmen etc may not be allowed to have access to the restricted area except for attending any rectification of defects pointed out to his by the GE.

4.10 **FIRE PRECAUTIONS**

4.10.1. The contractor, his agents, representative, workmen, etc shall strictly observe the orders pertaining to fire precautions prevailing within the restricted area. Motor transport vehicles, if any allowed by authorities to enter the restricted area, must be fitted with serviceable fire extinguishers.

5.0 **MINIMUM WAGES PAYABLE TO LABOUR**

5.1 Refer condition 58 of IAFW –2249. The contractor shall not pay wages lower than minimum wages for labour as fixed by Government of India/State Government/Union Territory whichever is higher.

5.2 Schedule of minimum fair wage is not enclosed along with the tender, but the contractor is deemed to have full knowledge regarding the minimum fair wages payable to the labour as legally effective on the day of submission of tender irrespective of the fact, the gazette notification may not have been published or that he was unaware of the increase in the minimum wages. For the purpose of special condition 26,the minimum wages legally effective on the date or receipt of tender shall be the basis for reimbursement, refund on variation in prices. The minimum wages legally effective referred above are the minimum wages notified in gazette or any local regulation whichever is higher.

5.3 The contractor shall have no claim whatsoever, if on account of local factor and regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

SPECIAL CONDITIONS (CONTD…)

6.0 **ROYALTIES**

6.1 Refer to condition 14 of General conditions of contracts ((IAFW –2249). No quarries on Defence land are available

7.0 **LAND FOR TEMPORARY WORKSHOP STORES ETC**

7.1 Delete the lines 5 to 9 of sub Para 1 of condition 24 of IAFW –2249 regarding in the event of the areas of land allotted to him and insert as under:-

7.2 Contractor shall be allotted free of charge the area as marked on the layout plans for the purpose of erection of temporary workshop, stores etc., No Ministry of Defence land is available for accommodation of labour and canteen for which the contractor shall make his own arrangements at his own expense.

8.0 **WATER**

8.1 Refer condition 31 of IAFW-2249. If contractor desires to buy water from Government, water will be supplied by the department at the point(s) as indicated in the site plan or as directed by GE. The water supply shall be metered and the contractor shall pay for after used in works and by his workmen at the rate of **Rs 3.75** per Rs 1000/- worth of work done. The supply of water may not be continuous, the contractor shall be deemed to have ascertained the hours of availability of water before submitting his tender. MES does not guarantee for break down in the system or if supply is not sufficient. The contractor shall make his own arrangements to store and supplement the water supply at his own cost.

8.2 The contractor if he so desires, will be allowed to augment the department's water supply by boring wells at his own cost at site(s) approved by the GE. On completion of the work, the contractor shall remove pipe etc. installed and make good the site as directed by Engineer-in-Charge without any extra cost of Government. No rebate shall be given for augmentation of water by the contractor and charges for water shall be recovered at the rate mentioned above. The contractor shall, however, ensure that water drawn from such sources is clean and free from all impurities and is suitable for mixing concrete, mortar, washing aggregate and curing of concrete as specified in IS 456..

9.0 **CO-OPERATING WITH OTHER AGENCIES**

9.1 The contractor shall permit free access and generally reasonable facilities to other agencies or department workmen engaged by Govt. to carryout their part of the work, if any under separate arrangements.

10.0 **ELECTRIC SUPPLY**

10.1 Electric supply required for the work shall be made available by MES at point [S] shown on site plan up to 50 KVA only. The exact location of the electric point [S] will be shown by the GE. KWH meters to register the electric energy supplied and main switch shall be provided and installed by the MES. Contractor shall provide all necessary cables, fittings, etc, from the tapping point in order to ensure a proper and suitable supply of electricity for execution of work. All contractor's installation shall conform to and strictly in accordance with Indian Electricity Rules. Moreover, the layout of cables etc as proposed by him shall be as per plan approved by the Engineer-in-Charge. The contractor will be charged for the electric energy consumed for execution of works at the following rates:

[a] **At Rs. 19.28 per unit for Lighting**.
[b] **At Rs. 19.28 per unit for Power**.

10.2 MES does not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for breakdown in the system. 9.3. GE or his representative shall be free to inspect all the power consuming devices or any electric lines provided by the contractor. Any devices or electric lines provided by the contractor, which is not to the satisfaction of GE, shall be disconnected from the supply, if so directed by him.

11.0 NET WORK ANALYSIS

11.1 The time and progress chart to be prepared as per condition 11 of General Conditions of contracts (IAFW –2249) shall consist of detailed net-work analysis and a time schedule. The critical path net-work will be drawn jointly by the GE and the contractor soon after acceptance of tender. The time scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. On completion of the time schedule a firm calendar date schedule will be prepared and submitted by the contractor to the GE who will approve it after due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over the site.

11.2 During the currency of the work, the contractor is expected to adhere to the time schedule and this adherence will be a part of the contractor's performance under the contract. During the execution of the work, the contractor is expected to participate in the reviews and updating the network under taken by the GE. These reviews may be undertaken at the discretion of the GE either as a periodical appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the time schedule as a result of the review will be submitted by the contractor to the GE within a week for his approval after due scrutiny.

11.3 The contractor shall adhere to the revised time schedule thereafter. In case of contractor disagreeing with revised schedule, the same will be referred to the accepting officer whose decision shall be final conclusive and binding. GE approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW –2249 and separately regulated.

11.4 Contractor shall mobilize and employ sufficient resource to achieve the detailed schedule, with in the broad frame work of the accepted method of working and safety. No additional payment will be made to contractor for any multiple shift work or other intensive methods contemplated by him in his schedule even though the time schedule is approved by the department.

12.0 SAMPLES OF MATERIALS

12.1 Refer condition 10 of IAFW –2249

12.2 Materials and articles shall bear IS certification marks and only these shall be incorporated in the work except for the materials for which specific makes are specified in tender.

12.3 The tenderer is advised to inspect other materials which are displayed in the office of GE before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials required to be incorporated in the work irrespective of whether he has actually inspected them or not. The materials to be incorporated in the work by the contractor shall be ISI marked or shall be superior specifications given hereinafter. The contractor shall not procure materials unless the samples are first got approved by the GE.

13.0 RECORD OF CONSUMPTION OF CEMENT

13.1 The contractor shall maintain a pucca bound register with serially numbered pages with all pages initialed by Engineer-in-Charge against numbering showing quantities of cement received, used in work and balance at the end of the day. The form of record shall be signed daily by representative of MES and the contractor in token of verification of its correctness and will be checked by Engineer-in-Charge at least one a week and on the days cement issued to the contractor.

13.2 The register shall be kept at site in safe custody of the contractor's representative during the progress of the work and shall be on demand be produced for verification to the inspecting officer(s).

13.3 On completion of the work the contractor shall deposit the cement register with the Engineer-in-Charge for record.

14.0 **ADVANCE ON ACCOUNT OF MATERIAL WHICH DOES NOT LOOSE IDENTITY (Condition 64 of IAFW-2249** – Advance on account)

14.1 Add the following in continuation of Para 8 of condition 64 of IAFW-2249 "Provided further the contractor may be paid advance on account of the full value of the materials brought on the site, on his furnishing guarantee bonds from a schedule bank for the amount of Retention Money which should otherwise be recoverable from him under the contract". Any non perishable materials which do not lose its identity, at the discretion of GE.

14.2 The Bank Guarantee Bonds shall be executed for a period and on a form as directed by the Accepting Officer. The contractor shall further arrange to extend the period of guarantee bond if and when necessary, as directed by the Accepting Officer or shall furnish fresh guarantee bond of similar value.

14.3 It may be noted that the advance on account to the full value of materials brought on the site is permissible only in respect of fittings and fixtures and other manufactured items which do not loose their identity. Materials like bricks, aggregate, cement, paints, Precast-concrete and similar items shall not be taken in the list.

15.0 **QUALIFIED TRADESMEN :**In compliance with the Condition 26 of IAFW-2249 (General Conditions of Contracts) the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from industrial Training Institute (ITI)/ National institute of Construction Management and Research (INCMAR)/ National Academy of Construction (NAC), Hyderabad/ Similar reputed and recognized Institutions by Stat/ Central Government, to execute the works or their respective trade. The number of such qualified tradesmen shall not be less than 25% of total skilled/ semi skilled tradesmen along with requisite certificates to Garrison Engineer for verification and approval. Notwithstanding the approval of such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades, leading to un-satisfactory workmanship, the Contractor shall remove such tradesmen within a week after written notice to this effect by the GE and shall engage other qualified tradesmen after prior approval of GE. GE's decision whether a particular tradesman possesses requisite qualification, skill and expertise commensurate with nature of work, shall be final and binding. No compensation whatsoever on this account shall be admissible.

16.0 **CONTRACOR'S REPRESENTATIVES AND WORKMEN**

16.1 Refer General condition 25 of IAFW-2249. The contractor shall employ only Indian Nationals as his representatives, servants and workman and verify their antecedents and loyalty before employment of them for the works. He shall ensure that no persons of doubtful antecedents and nationality in any manner shall be associated with work. If for the reason of technical collaboration or other consideration the employment of any foreign national is unavoidable the contractor shall furnish full particulars to this effect to the accepting officer at the time of submission of his tender.

SPECIAL CONDITIONS (CONTD…)

17.0 **SECURITY OF CLASSIFIED DOCUMENTS**

17.1 Contractor's special attention is drawn to conditions 2A and 3 of IAFW-2249 (General Conditions of Contractors). The contractor shall not communicate any classified information regarding the work either to sub contractor or other without prior approval of the Engineer-in-Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of the work and shall return all documents on completion of the work or earlier on termination of the contract.

17.2 The contractor shall along with the final bill attach a receipt of his having returned the classified documents as per condition 3 of IAFW-2249 (General Conditions of contract).

18.0 **RECORD OF MATERIALS**

18.1 The quantity of materials such as paints, bitumen, bituminous felt, water proofing compound, chemicals for anti termite treatment lead and the like as directed by the Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works) shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

18.2 Materials brought to site shall be stored as directed by the Engineer-in- Charge and those already recorded in measurement book shall be suitably marked for identification.

18.3 The contractor shall, on demand, produce to the GE original receipted vouchers / invoice in respect of the supplies vouchers/ invoice so produced and verified shall be stamped by Engineer-in-charge indicating contract number. The contractor shall ensure that the materials are brought to site, in original sealed containers / packing bearing manufacturer's marking except in the case of the requirement of materials being less than smallest packing.

18.4 Contractor shall produce vouchers / invoices from the manufacturers or their authorized agents for the full quantity of the materials as applicable as prerequisite before submitting claims for payment of advances on account of the work done and or materials collected in accordance with condition 64 of General Conditions of contracts IAFW-2249.

19.0 **SECURITY AGAINST LOSS OR DAMAGE**

19.1 The contractor shall furnish to the Engineer –in –charge every morning distribution /return of his plants / equipment on the site of work stating the following particulars:-

(i) Particulars of plants / equipment, their make, manufacturers model number if any, registration number if any, capacity, year of manufacturing and year of purchasing etc.,

(ii) Total number (quantity) on site of work.

(iii) Location, indication number, quantity at each location of site of work.

(iv) Purchase value on the date of purchase. For the purpose of the condition, plant/equipment shall be given vehicle number i.e., trucks but neither the workman tools nor any manually operated tools/equipment. The Engineer – in -charge shall record the particulars supplied by the contractor in the works diary and send the return to the GE for record in his office.

20.0 **OFFICIAL SECRET ACT:** The contractor shall be bound by the Official Secret Act 1951.

21.0 **CLEANING DOWN:** Refer condition 49 of IAFW –2249 General Conditions of contracts.

22.0 **VALIDITY OF TENDER:** The tender shall remain open for acceptance for a period of **60 (SIXTY)** days from the next date subsequent to last date of bid submission.

SPECIAL CONDITIONS (CONTD…)

23.0 DAMAGE TO OTHER STRUCTURES/BUILDINGS: Any damage caused to the existing structure [s] during the execution of the work or at any time during the currency of the contract shall be made good by the contractor at his cost to the entire satisfaction of the concerned authority and the site of work left clean and tidy on completion. Rectification/making good shall confirm to the standard of workmanship originally used in the work and finished work shall match the existing work in all respects to the entire satisfaction of the Engineer – in – Charge. In case of any dispute on this account the matter shall be referred to the Accepting Officer, whose decision in writing shall be final, conclusive and binding.

24.0 **CONCILIATION** : Refer condition 71 of IAFW - 2249

25.0 **REIMBURSEMENT/REFUND ON VARIATION IN PRICES:** The reimbursement/refund due to variation in prices of material, fuel and labour shall not be adjusted. The condition 63 of IAFW 2249 stands deleted.

26.0 **RE-IMBURSEMENT/REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TO CONTRACT VALUE:**

26.1 The rates quoted by the Contractor shall be deemed to be inclusive of all taxes / cesses viz GST, duties, Royalties, Octroi & other levies payable under the respective Statutes as applicable on the date of bid submission end date. No reimbursement/refund for variation in rates of taxes, duties, Royalties, Octroi & other levies, and/or imposition/abolition of any new/existing taxes/cesses, duties, Royalties, Octroi & other levies shall be made except as provided in sub paras 25 here-in-below: -

26.2 The taxes/cesses which are levied by Government at certain percentage rates of Contract Sum/Amount shall be termed as "taxes directly related to Contract value" such as GST, Labour Welfare Cess/tax and like as applicable but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all taxes directly related to Contract value" with existing percentage rates as prevailing on last due date for receipt of tenders. Any increase in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the Contractor and any decrease in percentage rates of "taxes" directly related to Contract value" with reference to prevailing rates on last due date for receipt of tenders shall be refunded by the Contractor to the Government/deducted by the Government from any payments due to the Contractor. Similarly imposition of any new "taxes directly related to Contract value" after the last due date for receipt of tenders shall be reimbursed to the Contractor and abolition of any "taxes directly related to Contract value" prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Government/deducted by the Government from the payments due to the Contractor.

26.3 The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to Contract value", give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating there to which he may be in a position to supply. The Contractors shall submit the other documentary proof/information as the GE may require.

26.4 The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Government, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other information as the GE may require.

SPECIAL CONDITIONS (CONTD…)

26.5 Reimbursement for increase in percentage rates/imposition of "taxes directly related to Contract value" shall be made only if the Contractor necessarily & properly pays additional "taxes directly related to Contract value" to the Government, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Government Authority and submits documentary proof for the same as the GE may require.

27 **EMPLOYEES PROVIDENT FUND:** Contractors shall be in a possession of EPF code number and all the workers employed by him shall be enrolled as a member of EPF and shall have EPF account number. The contractor shall render a certificate to the GE that all workers employed by him are registered with EPF.

28 **CONSTRUCTION LABOUR WELFARE TAX:** Consequent upon promulgation of ordinance by the President of India, Contract shall be liable to pay construction labour welfare tax / cess and quoted rates shall be deemed to be inclusive of this tax / cess as fixed by the Government and nothing extra shall be admissible to the contractor on this account.

SIGNATURE OF CONTRACTOR

AGE [CONTRACTS]
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATIONS SECTION- I

1.0 **GENERAL**

1.1 Work under this contract shall be carried out in accordance with Schedule 'A', particular specifications and general specifications and other provisions in MES Standard Schedule of Rates (here-in-after called MES Schedule) Part-I and Part-II read in conjunction with each other.

1.2 Term "General Specifications" referred to herein before as well as referred to in IAFW-2249 (General Conditions of contracts) shall mean the specifications contained in the MES Schedule Part-I.

1.3 General Rules, specifications, special conditions and all preambles in the MES Schedule shall be deemed to be applicable to the work under this contract, unless specifically stated otherwise in these documents in which case the provisions in these documents shall take precedence over the aforesaid provisions in the MES Schedule. The term 'as specified' wherever appears in tender documents and drawings, relates to relevant particular specifications and in its absence general specifications. All references to MES Schedule (Standard Schedule of Rates) in these specifications relate to Part I of MES Schedule unless otherwise mentioned. References to only some paragraphs of MES Schedule have been made in these particular specifications but other paragraphs and provisions as applicable are also to be followed for all sections/parts of Schedule 'A' even though not particularly mentioned hereinafter.

1.4 Particular Specifications given hereinafter shall be applicable to all the works covered in Schedule "A" and these are brief and only particularise and emphasize the specifications given in MES Schedule which are not repeated.

1.5 Where specifications/provisions given in these particular specifications are at variance with the provisions/specifications given in MES Schedule, specifications/provisions given in these particular specifications shall be followed. In case no specifications are given in these Particular Specifications for any item of work then the work shall be executed to the specification given in the MES Schedule in preference to the local trade practice. Where specification for any item of work is not given in these Particular Specifications or in MES Schedule in that case specifications given in relevant Indian standard specifications or code of practice shall be followed.

1.6 In the event of any discrepancy between various documents forming Part of the tender, the following order of precedence shall be observed:

[a] Description of Schedule "A" including notes therein.

[b] Particular Specifications [Sections I & II].

[c] MES Schedule [SSR].

1.7 **NOTES:** In case no provision is made in [a] to [c] above, Indian Standard Specification/Code of Practice shall be followed.

2.0 SAMPLES OF MATERIALS

2.1 The tenderer is advised to inspect samples of the materials which are displayed in the Office of the GE before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials he is required to incorporate in the Work irrespective of whether he actually inspected them or not. The materials to be supplied by the contractor shall conform to or be superior to the samples displayed and shall comply with the specifications given hereinafter.

2.2 The contractor shall produce samples of all materials (except proprietary materials for which provisions of Para 3 below shall be complied with) and obtain approval before making bulk order for their procurement and incorporation in the work. In respect, of materials for which samples are not kept in GE office or detailed specifications not given hereinafter the material shall comply with relevant latest Indian standard specifications and shall be of any Standard make.

3.0 PROPRIETORY MATERIALS

3.1 Proprietary materials such as water proofing compound, paint, bitumen and the like shall when brought to site will be inspected by the Engineer-in-charge. The quantity brought to site shall be recorded in measurements book (s) and signed by the contractor and Engineer-in-charge and check to ensure that the required quantity has been brought to site for incorporation in the work.

3.2 Proprietary materials brought to site shall be stored as directed by the Engineer-in-charge. The quantity already recorded in the measurement book (s) shall be suitably marked for identification.

3.3 The contractor shall on demand produce to the GE original receipted vouchers/invoices in respect of the supplies. The contractor shall ensure that the materials brought to site in original sealed container/packing; bearing manufacturer's marking except in the case of requirement of materials being less than the content of smallest packing.

4.0 LAYING OF INTERNAL ELECTRICAL LINES AND WATER SUPPLY LINES

4.1 Layout of internal electrical lines and water supply lines shall be marked by the contractor on walls, ceiling, roofs, floors etc., lightly with chalk or as directed for approval of the Engineer-in-charge before actually executing the work.

5.0 OBSERVANCE OF INDIAN ELECTRICITY ACT ETC.

5.1 The contractor shall be responsible for complying with the applicable provisions in the Indian Electricity Act/Rules.

5.2 The contractor shall employ properly qualified supervisors and skilled electrician holding proper license under the Indian Electricity Act and Rules. He shall, on demand by the Engineer-in-charge produce such evidence of qualification of his workmen/supervisors either at the time of entry or thereafter during the currency of contract. The provision shall be complied with even though the contractor happens to be enlisted for carrying out electrical work.

6.0 MAKING CHASES, HOLES ETC. ON WALLS AND OTHER SITUATION

6.1 Chase / holes etc., shall be left in concrete, brick work, stone masonry floors and in any other situations for carrying out the various items of work as required for as directed by the Engineer-in-charge as far as possible during the progress of work in the particular situation and shall be made good on the same mortar / concrete as specified for the portion of the work.

7.0 **SITE CLEARANCE**

7.1 Refer condition 49 of IAFW-2249, General conditions of contracts. The contractor shall remove from the site all unused stores and materials, tools, plants equipment scaffolding temporary buildings, huts and the like belonging to the contractor provided for the execution of work under this contract and the site of work shall be cleared off all waste materials by the contractor and deliver the site in neat. Clean and tidy manner to the satisfaction of the Engineer-in-Charge on or before the data of completion. Nothing extra whatsoever shall be paid to the contractor for such clearance of site.

8.0 **CLEANING DOWN**

8.1 Refer condition 49 of IAFW-2249, General conditions of contracts. The contractor shall clean floors remove cement/lime paint drops etc., clean the joinery glass panes etc., and touch up all paints work in connection herewith and leave the whole premises clean and tidy to the entire satisfaction of the Engineer-in-charge before handing over the building.

9.0 **INDIAN STANDARD SPECIFICATIONS**

9.1 Materials provided by the contractor for incorporation in the works shall unless otherwise specified in particular specifications comply with the requirements of the relevant Indian Standard Specifications.

10.0 **APPROVED NAMES FOR PRODUCTS TO BE INCORPORATED IN WORKS:** The contractor can procure any one of the maker of materials as specified in Schedule for incorporation in the work. All fittings wherever available shall be ISI marked and preferred. If ISI marked fittings are not available those shall be conformed to ISI and shall be procured after approval.

SIGNATURE OF CONTRACTOR

AGE [CONTRACTS]
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATIONS SECTION- II

1.0 **GENERAL**

1.1 General rules, specifications for materials, workmanship and standard specification, referred to the MES standard Scheduled of Rates 2009 Part I and 2020 Part II (here-in-after referred to as MES Schedule of SSR) shall be deemed to apply to this contract unless mentioned otherwise in these tender documents. Where these are at variance, the later shall take precedence.

1.2 Materials to be provided by the contractor for incorporation in the work shall unless otherwise specified in the particular specification comply with their requirements of the relevant Indian Standard (IS) of the year of publication (Edition/Revision) as specified in MES schedule 2009 (Part I) or published on a data prior to issue of tender whichever is later. The contractor may, however, at his own discretion provide and incorporate in the works such materials complying with the requirement of any subsequently amended or revised edition of the corresponding IS without any price adjustment of the lump sum or while pricing deviation.

1.3. Particular specifications given here-in-after are in brief and intended only to particularize, amend and emphasis the specification given in MES Schedule which are not repeated here-in-after.

1.4. Specification of materials and workmanship shall be all as directed in MES Schedule except where specifically mentioned otherwise in these documents. General rules, preambles to various section/rates, special condition, method of measurement given in MES Schedule shall also apply to the contract unless otherwise mentioned in the tender documents. The workmanship must conform to the best engineering practices.

2.0 **SCOPE OF WORK**

2.1 The work under this contract shall be carried out in accordance with Schedule 'A', the particular specifications and other provisions in MES schedule. The work shall be carried out under the jurisdiction of AGE B/R SUNABEDA.

3.0 **SITE OF WORK**

3.1 The Works shall be carried out at different Locations/Site(s). The Contractor shall finalise the programme with the Engineer-in-Charge well in advance in such a way that neither the USERS feel inconvenience nor Work is delayed.

3.2 Buildings under occupation of USERS where Works to be executed, the Contractor shall made himself aware about the Nature of the Work involved for carrying out Works against the Items catered in SCHEDULE 'A'. The Contractor is deemed to have inspected the Site of Work and nothing extra shall be admissible to Contractor on this ground.

3.4 **VISIT OF SITE**

3.4.1 The Tenderers are advised to Contact GE on any Working Day during Working Hours, who will arrange for their Visit to Site of Work.

3.4.2 The Tenderer shall be deemed to have visited the Site of Work before quoting their UNIT RATES and no Claim for any EXTRA PAYMENT on account of any misunderstanding, etc. will be entertained by the Department, irrespective of the fact whether they have actually visited the Site or not.

4.0 **LAYOUTS**

4.1 Exact Layout of Pipe Line, Cable, etc. and Accessories, etc. shall be as directed by Engineer-in-Charge to suit/meet the Site Requirements. No Adjustment in the quoted Unit Rates, shall be made on account of such variations in Layout.

5.0 **STANDARD OF QUALITY & WORKSMANSHIP**

5.1 **GENERAL**

5.1.1 The Work shall strictly comply with the provisions contained in the Latest Edition of INDIA STANDARD CODE OF PRACTICE and/or IS SPECIFICATION as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

5.2. **MATERIALS**

5.2.1 All materials to be supplied by contractor for incorporation in the work shall conform to relevant specifications. In case specification for any materials needed for incorporation in the work is not contained in any of the contract documents the specification of such materials proposed to be incorporated in work shall be got approved in drawing from the GE before incorporation in the work.

5.2.2 All manufactured materials/articles other than those manufactured in contractor's workshop at site of works shall bear IS certification marks. In case any such articles do not bear the IS certification marks but conform to relevant specifications and are proposed to be incorporated in the works, samples of the same shall as per samples kept in GE's officer and shall be got approved in writing from the GE before their incorporation in the works. The contractor shall submit sufficient evidence to Engineer-in-Charge to show that such articles conform to the relevant specifications.

5.2.3 The contractor shall supply samples of materials taking into account delivery period (s) after placing firm order and allowing ample time for their testing and approval in order that if the samples supplied fail to amply with the specified requirements there will be sufficient time for him to submit fresh samples well before the materials are required to be used.

5.2.4 All manufactured materials/articles shall be brought at site in original sealed containers/packings bearing manufacturer's marking unless the quantity required is a fraction of the smallest commercial packing. Sealed container shall be opened in the presence of Engineer-in-Charge as and when required and empty container should be kept in contractor's store till the completion of the project for inspection by GE/other Inspecting Officers.

5.2.5 Tenderers attention is drawn to conditions 10A of IAFW-2249 (General Conditions of contracts) as amended vide amendment No 23. The contractor shall comply submission of proof of procurement of materials, as detailed therein. However, the purchase Vouchers in respect of following items is also required to be submitted irrespective of its value:-

(i) Cement, Steel

(ii) Materials of proprietary nature such as paints, water proofing compound, bitumen for water proofing of the roof, chemicals for anti-termite treatment and the like, quantity of which can not be ascertained after their incorporation in the works, shall be measured and recorded in the measurement books as soon as the same is/are brought to site and approved for use. Such measurements will be signed by Engineer-in-Charge and the authorized representative of the contractor. The contractor is required to obtain all such materials direct from the manufacturers/ their authorized dealers (if any) and to submit to Engineer-in Charge original stamped receipted bills along with relevant invoices in support of having brought the full quantity required for incorporation in the work and get these documents duly endorsed by the Engineer-in-Charge to the effect that the same materials and quantity have been used in the works. A record of their use shall be kept date wise in works dairy. The contractor shall comply with such instructions as are given to him in writing from time to time by the Engineer-in-Charge.

(iii) The contractor is required to submit purchase vouchers in respect of all items for which payment is claimed through RARs, as materials lying at site.

5.2.6 All building materials like aggregate, sand, bricks etc shall generally conforming to the samples kept in GE's office in addition to their conforming with relevant specifications given in the tender documents. The samples of all such materials shall be got approved from the GE in writing before these materials are brought to site in bulk. In case of any difference between samples kept in GE's office and as that specified in respective IS the specification laid down in IS shall take precedence. In case any difference in option between GE and contractor in this regard, the decision of CWE shall be final and binding. The sources of some of the materials are indicated below:-

5.2.7 Letters conveying approval of samples of materials by GE shall interalia mention sources of supply, name of manufacturers, traders name/brand (if any) and reference to clause of tender documents containing specifications of particular materials.

5.2.8 Contractor at his own may procure materials from other sources without any extra cost to the Deptt provided that the same conform to respective IS as stated here-in-before.

5.3 **Testing of materials** :-

5.3.1 Before incorporation in the work, the building materials, concrete etc are required to be tested for quality control in material testing laboratory.

5.3.2 The contractor may be permitted to set up site Laboratory to carry out necessary tests of materials. The type of test to be carried out shall be as directed by GE. Contractor shall be deemed to have quoted accordingly. No claim whatsoever shall be entertained at a later date for setting up site Laboratory and testing. However, contractor will be allowed to take out all the testing equipments after satisfactory completion of work and leave the site neat and tidy. In case contractor does not set up the site Laboratory, the requisite tests shall either be carried out in MES Zonal Laboratory (if such facilities exist) shall be carried out at the Laboratory mentioned in PS Clause 5.3.3 here-in-after at his own arrangement and expenses.

5.3.3 However, a percentage/selected tests, as directed by the Accepting Officer, shall be got done independently in Zonal Laboratory of CE Kolkata Zone and recovery charge for these tests shall be effected from the contractor. If no such tests facility exists in Zonal Laboratory, the same shall be carried out at the following Laboratories/Institutions at the contractor's own arrangement and expenses :-

a)	National Test House, Alipore, Kolkata.	b)	Jadavpur Engineering College, Kolkata
c)	SEMT Pune.	d)	Regional Testing Laboratory, 110/111 BT Road, Kolkata.
e)	Bengal Engineering College, Shivpur.	f)	Regional Engineering College, Durgapur.

5.3.5 In case any test is not conducted in above said test house/college, the test should be carried out in any Govt. laboratory / Govt. approved laboratory/Institution as approved by GE.

5.3.6 The contractor shall supply all the materials, moulds etc required for testing including transportation of materials without any extra cost.

5.3.7 Contractor shall employ competent technical person as approved by the GE to carry out tests in the site laboratory established by him. Contractor's quoted rate shall be deemed to have included the expenses of employment of technical person.

5.4 **WORKMANSHIP**

5.4.1 Work shall be carried out all as per provisions specified in SSR part I and IEE rules. The work shall be carried out in a workman like manner by trained personnel in the trade. The contractor is required to produce proof regarding qualification and experience of each tradesman employed by him to the Engineer-in-Charge on demand. Damages caused to any installation as a result of faulty workmanship or due to negligence, in the opinion of the Accepting Officer, the contractor shall have to make good such damages at his own cost, for which no claim what-so-ever is admissible.

6.0 **CEMENT TO BE SUPPLIED BY THE CONTRACTOR**

60.1 Full quantity of cement required for this work shall be supplied by the contractor under his own arrangement. Ordinary Portland cement of Grade 43 or higher conforming to IS:269-2015 (latest revision) or Portland Pozzolana Cement conforming to IS:1489-1991 shall be incorporated in the work. Clause-4.3 of SSR Part-I, 2009 may also please be referred.

Note: - In case more than one type of cement is used in any work, a record shall be kept showing the location and the type of cement used. Different types of cement shall not be mixed together.

6.2 **PROCUREMENT**: -

6.2.1 Cement supplied by the Contractor shall be procured from the main producers or from their authorized dealers/stockists. The following are the main approved producers of cement :-

Sl	Company Name	Brand	Type of Cement
1)	The Associated Cement Companies Ltd	ACC	-All-
2)	Ultra Tech Cement Ltd	ULTRATECH	-All-
3)	The India Cement Ltd	-	-All-
4)	M/S Dalmia Cement (Bharat) Ltd	DALMIYA / KONARK	-All-
5)	Century Cements	CENTURY	-All-
6)	Saurashtra Cement	SAURASHTRA	-All-
7)	The Ramco Cements Ltd	RAMCO	-All-
8)	Mangalam Cement Ltd	MANGALAM	-All-
9)	Birla Corporation Ltd	BIRLA	-All-
10)	Orient Cement	ORIENT	-All-
11)	Nuvoco Vistas Corporation ltd	NUVOCO	-All-
12)	Shree Cement	SHREE	-All-
13)	JK Cement	JK	-All-
14)	JK Lakshmi Cement Ltd	JK LAKSHMI	-All-
15)	Jaypee Rewa Cement	JAYPEE	-All-
16)	Ambuja Cement Ltd	AMBUJA	-All-
17)	M/S Shree Guru Kripa Cement (Pvt) Ltd	SARTAJ	OPC 43 Gr / PPC
18)	M/S Parasakti Cements Ltd	PRASAKTI	OPC 43 Gr / PPC
19)	M/S My Home Industries	MAHA CEMENT	OPC 43 Gr / PPC/PSC
20)	M/S Chettinad Cement	CHETTINAD CEMENT	OPC 43 Gr / PPC
21)	M/S Sanghi Industries Ltd	SANGHI	OPC 53 Gr / PPC
22)	M/S Wonder Cement Ltd	WONDER CEMENT	OPC 43 Gr / OPC 53 Gr / PPC
23)	M/S Kesoram Industries Ltd	BIRLA SHAKTI	OPC 43 Gr / OPC 53 Gr / PPC
24)	M/S JSW Cement Ltd	JSW PSC	PSC

6.3 Following conditions shall have to be satisfied before PPC will be allowed to be used in work:-

a) PPC should meet the strength criteria of 43 Grade OPC as laid down in IS : 269 – 2015.

b) Stripping time shall be 14 days.

c) In cold climate regions where temperature is lower than 15° C and important structures like overhead reservoirs, underground sumps, prestigious buildings and buildings with spans 10 metres or more, only OPC shall be used.

d) Both OPC & PPC shall not be permitted for use in the same building except for plaster & mortar.

e) While procuring PPC and PPC shall, following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC :-

i) The quantity of fly ash is strictly as per IS -1489 (Part-I) – 2002.

ii) Fly ash is inter ground with clinker not mixed with clinker.

iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping should be used.

iv) The fly ash received from thermal power plants using high temperature combustion above 1000 degree C should be used.

v) The fly ash content in PPC shall not exceed 25% to ensure consistency.

6.4 The particulars of the manufacturer /main producer of cement along with the date of manufacturer shall be submitted by the contractor for every lot of cement separately. The documents in support of purchase of cement will be verified by the site staff and GE.

6.5 **STORAGE OF CEMENT:**

6.5.1 There will be two separate godowns provided by contractor at site of works exclusively for storage of cement:-

(a) " First Godown" shall have the cement lot(s) for which independent testing is to got done and/or under process.

(b) " Second Godown" shall have the cement lot(s) whose independent testing had been got done and after successful testing stands approved by GE/AGE (I) for incorporation in the work.

6.5.2 Cement shall be stored over dry platform with gap of 20 cm room walls all-round and at least 20 cm high from floor level. The stacking of cement shall not be more than 10 Bags high. Compliance of provisions in clause 4.3.1, 4.3.2, and 4.3.3 of MES Schedule Part I shall be ensured. Godowns shall have easy access for proper inspection & cement shall be protected from dampness/moisture to minimize storage deterioration or intrusion of foreign matter. **Cement older than Six months shall not be used in RCC works.**

6.5.3 Both the Godowns shall be provided with two locks on each door. The key of one lock at each door shall remain with the Engineer-in-Charge or representative and that of the other lock with the contractors authorized agent at the site of works so that cement is removed form the godown only according to the requirements for independent testing purposes or for daily consumption purposes with the knowledge of both the parties.

6.6 **TESTING OF CEMENT:** Contractor shall submit particulars of manufacturer of cement along with the date of manufacture for every lot of cement separately. The manufacturer is to carry out inspections and testing of cement in accordance with the relevant BIS provisions. Contractor shall also submit manufacture's test certificates in original along with the test sheet or an authenticated copy thereof giving results of each physical test as applicable and the chemical composition of cement. The documents giving above particulars shall be verified by site staff and GE/AGE (I) in support of the purchases of cement and these details shall be recorded in the cement Acceptance Register as per Performa given in Appendix 'A' to the tender documents here in after.

6.6.1 The GE shall organize independent testing of random samples of cement drawn from the various lots which shall be got tested by the contractor under his own arrangements and the cost of both materials and labour required for sampling, testing, packing, transportation of samples to the testing places and testing charges thereof, and for subsequent identification of cement sampled, shall be borne by the contractor and is deemed to be included in the rates quoted by the contractor. The independent testing shall be got done from any of the following where facilities exist for all the tests required, viz, National Test House, Calcutta, National Test House, SEMT Wing – CME, Pune, Regional Research Laboratories, Zonal Laboratories, Govt Engineering College and Govt approved laboratories.

6.6.2 The independent testing shall be got done as per: -

(a) IS-5335 of 1986 : Method of sampling hydraulic cement.

(b) IS-4031 : Methods of physical test for hydraulic cement.

(c) IS-4032 of 1985 : Method of Chemical analysis of hydraulic cement.

6.6.3 The other relevant IS shall also be followed. However the tests shall be restricted to those indicated here in below: -

Srl No	Materials	Test	Method of Testing	Frequency of tests	Remarks
1.	Cement	(a) Setting time	IS-4031-Part-5 1988 REAFFIRMED – 2000	Once for each consignment or as and where required	
		(b) Soundness	-do- Part-3	-do-	
		(c) Compressive Strength	-do- Part-6	-do-	
		(d) Fineness	IS-4031 Part-2 1999	-do-	

6.6.4 The above tests shall be independent tests. The samples shall be taken immediately within 3 weeks of delivery at the site of works and all the tests got carried out by contractor within one week of sampling. In case it is not possible to test the samples within one week, the samples shall be packed and stores in airtight containers as decided by GE/AGE (I) till such time that they are tested.

6.6.5 The entire lot(s) of cement shall stand rejected, if it does not comply with any of the requirement of IS specifications and the test results on samples are not within the acceptable limits. The rejected lot(s) shall be removed from the site of works by the contractor under their own arrangement and cost. The cost of tests shall be borne by the contractor irrespective of the results.

6.6.6 The contractor shall use only one type brand of cement for entire structural /reinforced concrete work. In the event of contractor desiring use of any different brand of cement, fresh design mix shall be carried out and submitted for approval without any extra cost to the Govt.

6.7 **DOCUMENTATION:**

The contractor shall submit original paid vouchers from the manufactures for the total quantity of cement supplied under each consignment brought to the site of work.

6.7.1 All consignments so received at the work site shall be inspected by GE along with the relevant documents for permitting storage in the 'first godown' defined here-in-before.

6.7.2 The original vouchers and the test certificates shall be kept In record in the office of GE duly authenticated and with cross reference to the Control Number recorded in the Cement Register as per format given in Appendix-A here-in after, maintained for "first godown" duly signed by JE(Civil), Engineer-in-Charge, GE and the contractor.

6.7.3 After independent testing has been got done by GE and satisfactory results obtained and kept on record the consignment so tested shall be removed from the 'first godown' and transferred to the 'second godown' as defined here-in before. Original vouchers and the test certificates submitted by the contractor along with the consignment and test certificates of independent testing shall be defaced by Engineer-in-Charge and kept on record duly authenticated and with cross reference to the control Number rendered in the cement register maintained for the "Godwon" This register shall be termed as "Cement Acceptance Register" which shall be signed by JE(Civil) , Engineer-in-Charge, GE and the contractor. Besides this the quantity of cement shall be also suitably recorded in the Measurement Book for record purposes before incorporation in the work and shall be signed by Engineer-in-Charge and the contractor.

6.7.4 The Accepting Officer may order a Board of Officers in which contractor shall also be a member for random check of cement and verification of connected documents.

6.7.5 Daily inspection shall be carried out by Engineer-in-Charge or his representative and contractor's authorized agent and separate register shall be maintained duly indicating on daily basis the inspection carried out and indicating holding in godown at beginning of the day, inward supplies in the day, removal in the day and balance at the close of the day duly signed by above persons. In the removal column particulars of lot removed and purpose for which removed shall be indicated.

6.7.6 The purchase voucher and test certificate of manufacturer shall be linked.

6.8 **WEIGHING AND PAYMENT FOR CEMENT**.

6.8.1 Random sample of 5 bags per 100 bags shall be got weighed by Engineer-in-Charge in presence of Representative of contractor from each consignment to arrive at average weight per bag for calculating total quantity of cement bought for incorporation in the work. All arrangement for weighing of cement bags including labour etc. shall be provided by the contractor at their own cost.

6.8.2 The quantity of cement arrived at as aforesaid shall be considered for payment as applicable in accordance with IAFW-2249, General Conditions of Contracts.

6.8.3 Only when the rest results of independent testes carried out on the cement are found satisfactory and it is permitted to be stored in the 'Second Godown' and quantity entered in MB, then only this lot shall be considered for payment in RARs.

6.8.4 Payments shall be restricted as per provisions in clause No 64 "Advance on Account: of General Conditions of Contracts (IAFW-2249).

7.0 **CONCRETE WORK**

7.1 Coarse and fine aggregate for cement concrete: Refer paras 4.4.1 to 4.4.6 of MES Schedule Part-I. The coarse aggregate shall be crushed stone aggregate and fine aggregate shall be sand. Use of sand conforming to grading Zone-IV of IS 383 shall not be allowed in RCC/PCC work.

7.1.1 Grading of coarse aggregate for plain and reinforce concrete i.e. stone aggregate shall be as per Para 4.4.7.1 of MES Schedule Part I.

7.1.2 Fine aggregate shall be natural sand or crushed stone free from admixture of clay, loam, silt, organic matter and other impurities and shall conform to Zone II of table IV of IS-338. The aggregate shall be drawn from approved quarries.

7.1.2 All cement concrete, shall be mixed in mechanical mixer. However, in case of small quantity, the contractor may obtain written permission of Engineer-in-Charge, to adopt hand mixing over watertight platform and care shall be taken to ensure mixing is uniform in colour and consistency.

7.1.2 Samples of the aggregates proposed to be used shall be approved by the Accepting Officer, prior to bulk delivery to the site of work. Field tests for determining the contents of silt, loam, clay etc for fine aggregate shall be carried out by the Engineer-in-Charge from time to time to ensure that materials brought to site are inconformity with the samples approved by the Accepting Officer.

7.2 **Water proofing compound:** Refer Para 4.8 of MES Schedule Part I.

7.3 **Mix & grade of cement concrete**: Refer Para 4.11.1, 4.11.2 and 4.11.2.1 to 4.11.2.4 of MES Sch Part I.

7.4 **Workability of concrete**: Refer to Para 4.11.4 of MES Schedule Part I & Para 7 of IS-456-2000.

7.5 **Mixing of concrete**: Refer to para 4.11.5 and 4.11.5.1 of MES Schedule Part I.

7.6 **Transportation of concrete**: Refer to paras 4.11.9 of MES Schedule part I.

7.7 **Placing of concrete**: Refer para 4.11.10. to 4.11.10.3 of MES Schedule part I.

7.8 **Compaction of concrete**: Refer Para 4.11.11 of MES Schedule Part I. Compaction of concrete in RCC slab, beam walls and columns shall be done with approved mechanical vibrator. In other locations hand compaction may be permitted by taping or Roding as approved by the GE.

7.9 **Curing of concrete**: Refer to Para 4.11.13 of MES Schedule Part I.

7.10 **Protection and Inspection of Concrete**: Refer paras 4.11.14 and 4.11.15 of MES SSR part-I.

7.11 **Concrete covers to reinforcement**:Refer to paras 4.11.8.1 to 4.11.8.4 of MES Schedule Part-I.

8.0 **FORM WORK** : Refer to Para 7.15 of MES Sch Part I. Unless otherwise specified under relevant item of BOQ, form work shall be for rough finish. Camber shall be provided as per direction of Engr-in-Charge.

9. **CURING & MINIMUM PERIOD FOR STRIKING OFF FORM WORK WITH POZZOLANA CEMENT USED.**

9.1 Where pozzalana cement is used for RCC works in various locations, the minimum period for curing and striking off form work shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Part-I.

9.2 **Curing**: -

(i) Structural RCC Work : 21 days

(ii) PCC Work, Plastering Etc. : 14 days

9.3 **Striking off form work**: -

(i)	Walls ; columns and vertical sides of beams	: 3 to 4 days
(ii)	Slabs(Props left under)	: 8 Days
(iii)	Beams soffits(Props left under)	: 9 Days
(iv)	Removal of props to slab/beam:-	
	(aa) Spanning up to 6M Span	: 16 days7
	(ab) Spanning beyond 6M Span	: 22 days

(v) For cantilever portion of slab, form work with support shall be retained until the completion of the casting of the entire frame work of the building.

9.12.4 In case of bad weather, periods mentioned above may be revised at the discretion of the Engineer-in - Charge. The contractor shall be deemed to have considered the above provision before tendering and quote rate accordingly. No claim will be entertained if longer periods required for striking off form work and curing and all such effected matters is looked into upon the use of pozzolana cement as a result being instructions issued to the contractor in full or part of for bad weather.

9.12.5 Curing shall be carried out using pump of suitable capacity at no extra cost to the Govt all as directed by Engineer-in-Charge.

10.0 **FLY ASH CEMENT BRICKS**

10.1 Fly ash cement bricks (here after termed as bricks) shall conform to the requirements laid down in IS 13757-1993 and IS 12894 – 2002. Bricks shall have minimum compressive strength 75 Kg/CM2 (Class designation 7.5).

10.2 The brick shall have smooth rectangular faces with sharp corner and slight round edges, and shall be well burnt, uniform in colour, free from cracks, flaws, nodules of lime and emit clear ringing sound when struck.

10.3 Type and size of Bricks shall be as under :-

10.3.1 Type of bricks - Non modular bricks

10.3.2 Size of bricks - 250 x 125 x 75mm

10.3.3 Dimensional Tolerance for group of 20 bricks :-

Length 20 x Length of Bricks + 80mm

Width 20 x width of Bricks + 40mm

Height 20 x height of Bricks + 40mm

10.4 **SAMPLING & TESTING** – Sampling & Testing of Bricks shall be carried out by the Contractor as per IS-5454 and IS-3495 (Part-I to Part-IV) and as directed by GE without any Extra Cost to the Department.

10.5 **CEMENT**

10.6 Cement shall be as specified in CONCRETE WORK here-in-before.

10.7 **SAND**

10.8 Refer Clause No. 5.4 and 5.4.1 of MES Schedule Part-I. Sand for the Work shall be naturally occurring River Bed Sand and conforming to Sample kept in GE's Office and/or as approved by GE. Sand shall not contain any type of harmful Materials, shall be free from all type of Impurities, Dust, Silts, etc.

10.9 Sand shall be stored at Site in Bins or in Dumps over Brick Platform or by using any other approved method so as to avoid contamination or shoveling all as directed.

10.10 **WATER**

10.11 Refer Clause No. 5.18 of MES Schedule Part-I.

10.12 **WORKMANSHIP**

10.13 **GENERAL** – Refer Clause No. 5.19.1, 5.19.2, 5.19.6, 5.19.7 and 5.20 to 5.32 of MES Schedule Part-I. During execution of Work, Centre Line Dimensions shall be maintained.

10.14 **HALF BRICK WALLS**

10.15 Refer Clause No. 5.37 of MES Schedule Part-I and shall be constructed in Stretcher Bond.

10.16 The Width of the Bricks for Half Brick walls shall be considered as 110mm and deviation if any shall not be made for Bricks used. Centre Line dimensions shall, however be maintained in case of variation in size of Bricks from that specified in Tender. No adjustment of Extra Cost due to increase in width of Foundation Concrete, Walls, Lintels, Band & Blocks etc. shall be admissible to Contractor on this account.

11.0 **WOOD WORK (CARPENTERS WORK AND JOINERY)**

11.1 Form work: Refer to Para 7.15 of MES Schedule Part I. The formwork shall be for rough finish. No chamber in form work as mentioned in para 7.15.7 of MES Schedule Part I need be provided.

11.2 Timber and defects in timber: Refer to para 7.3.1,7.3.1.1,7.4.1,7.4.2,7.5.1 & 7.5.2 of MES Sch Part I.

11.3 Timber for wooden work and joinery (Except particle board and block board) shall be of second class hardwood as mentioned in Schedule 'A' and the following species: -

(i)	Timber for fillets work etc.	-	Second class hard wood of Hollock/ Panisaj / Kalasiris / Chickrashi.
(ii)	Timber for door / window frame	-	Second class hard wood Sal.
(iii)	Timber for scantling work etc.	-	Second class hard wood of Siris / Gamari/ Panisaj / Kalasiris

11.4 Workmanship General: refer para 8.15 of MES Schedule Part-I. Unless otherwise specified in Sch 'A' all exposed faces of woodwork and joinery shall be wrought. Unexposed surfaces shall be clean sawn.

11.5 **SEASONING**

11.5.1 Seasoning and moisture contents: refer to para 7.6 and 7.7 of MES Schedule Part I. Moisture contents shall be as specified for Zone IV.

11.5.2 All timbers shall be kiln season by suitable process conforming to IS-1141-1973 (Code of practice seasoning of timber) to maintain moisture content as permissible tabulated in clause 7.7 of SSR Part-I (2009) and in IS-287-1973 (recommendations of maximum permissible moisture content for timber used for different purpose). Seasoning shall be carried out in two stages i.e. to remove partial moisture content before treatment and then to maintain desired percentage of moisture content after treatment.

11.6 **TREATMENT OF TIMBER**

11.6.1 All timber incorporated in the work to be manufactured at site shall be treated as mentioned in IS 401 by modified hot and cold treatment, the minimum quantity of chemical/impregnated in each cubic metre of timber shall be at the rate as specified in Table 2 Group 5 of the IS. Cost of treatment to woodwork shall be deemed to be included in the cost of respective item of Sch 'A'.

11.6.2 Rates of Schedule 'A' for items of woodwork and joinery shall be deemed to include above preservation treatment.

NOTES: -

1. The above treatment is not applicable to particle board/plywood and to teak wood members, which shall be treated as specified in relevant IS.
2. The composition & concentration of chemical s shall be as per Clause 3.42 & 3.43 of IS-401.
3. ASCU preservation will be accepted against copper chrome arsenic composition without any price adjustment.
4. The contractor may, however use factory made pressure treated timber with prior approval of GE in writing without any price adjustment. The contractor shall produce certificate from the factory for such treatment.

12. **JOINERY**

12.1 **PANELLED,GLAZED AND GAUZED SHUTTERS.**

12.1.1 Refer Para 8.21.1 to 8.21.7 & 8.24 of SSR Part-I. Panelled, glazed, gauged shutters shall be factory made.

12.2 **FLUSH DOOR SHUTTERS**

12.2.1 Refer para 8.25 and ,8.25.1 to 8.25.9 of MES Schedule Part-I. Flush door shutters shall be ISI Marked [IS: 2202 (Part-I) – 1999.

12.3 **PVC DOOR AND FRAMES**

12.3.1 Irrespective of whatever shown on drawings, toilet door frames shall be factory made solid PVC door frames. The same shall be of size 50 mm x 47 mm made out of 5 mm PVC sheet. The door profiles are to be reinforced with 19 x 19 mm MS square tube. EDPM rubber gasket weather seal shall be provided throughout the frame. The profile shall be mitre cut at two corners & joined by inserting 2 nos. 150 mm long brackets of 15 x 15 mm square tube inside 19 mm x 19 mm M S square tube & suitable screwed.

12.3.2 Door frames for moulded door shutters shall be of size 100mm x 45mm with a wall thickness of 5mm ($\pm$0.25)

12.3.3 Irrespective of whatever shown on drawings, toilet door shutters shall be PVC panelled shutter with styles, top & bottom rails of size 90 x 37 mm & lock rail of size 105 x 37 mm with multi cavity hollow sections 2 mm average wall thickness all as shown at detail in drg No. AF/TD/142 sheet 1/1. The panel inserts shall be multi cavity hollow sections of size 100 x 20 mm with wall thickness of 1.2 mm. The corners of styles & rails shall be mitred cut & fixed with solid PVC 'L' shaped brackets two nos of size (200 mm x 100 mm & 100 mm x 100mm) at each corners. Two nos 6 mm dia MS rods with two coats of red oxide primer shall be inserted horizontally at appropriate levels through styles & panel inserts with check nuts and PVC washers. The door shall be as approved by GE and fixing shall be done as directed by Engineer-in-Charge.

12.4 **ALUMINIUM GLAZED DOOR / WINDOW / PARTITION**

12.4.1 Unless directed otherwise, aluminium doors, windows, ventilators, partitions, composite units etc and fittings shall be anodized and minimum average thickness of anodizing (coating of anodizing) on all aluminium section and fitting shall be 15 micron and shall conform to IS 1868-1996. Testing of anodizing coating shall be as per IS 1948-1961.

12.4.2 Thickness of Powder Coating (of any colour) wherever indicated, shall be not less than 50 micron.

12.4.3 Jointing of sections, providing fittings, lugs, method of fixing etc shall be as per IS 1948-1961.

12.4.4 Aluminium doors / windows / ventilators / partition shall be made in accordance with Para 10.37 of SSR Part I 2009 under the close supervision of the Engineer-in-Charge. The dimension of members shall be specified in SSR. Unless otherwise directed, all panels shall be of anodized / powder coated aluminium panel, cut to required size with minimum 12 mm portion being inserted in the frame and shall be fixed firmly with CP brass or stainless steel screws. The joints between panels and members shall be sealed with epoxy resin or silicon sealant. Joints shall be filled with PVC/ Neoprene felt, cleats as directed by the Engineer-in-Charge.

12.5 **ALUMINIUM GRILL**

12.5.1 The aluminum grill shall be provided from approved manufacturer . Design of aluminum grill shall be all as shown on drawings/as approved by GE and shall be fixed to the windows or other locations as per manufacturer specification as approved by GE.

13.0 **BUILDER'S HARDWARE**

13.1 GENERAL : Articles of builder's hardware (fittings) shall be of aluminium alloy. The type and size of fittings shall also be as indicated. Fittings shall be of approved make. Refer para 9.2.1, 9.2.2, 9.2.3, 9.2.4 & 9.2.5 of SSR Part I.

13.2 Screws – Refer para 9.2.6 of SSR Part I.

13.3 Tower bolt - Refer para 9.3 of MES Schedule Part-I. The diameter of bolts shall be 10 mm and the bolts shall be barrel bolts.

13.4 Butt hinges - Refer para 9.7 of SSR Part- I. Butt hinges shall be of pressed steel/cold rolled mild steel medium weight.

13.5 Ball catch shall be of cast brass all as specified in para.9.14 of SSR Part-I.

13.6 All builder's hardware shall be ISI marked except bow handles which shall conform to relevant IS.

13.7 Mortice Lock – Refer para 9.9 and subsequent paras thereof. Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

13.8 Handles - Refer para 9.11 and subsequent paras thereof . Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

13.9 Floor Door stoppers - Refer para 9.15 and subsequent paras thereof . Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

13.10 Door Closers - Refer para 9.16 and subsequent paras thereof. Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

13.11 Wire cloth for mosquito proofing shall be stainless steel wire cloth 0.36 mm nominal dia. of wire and average width of aperture 1.40 mm fixed with stainless steel wire staples.

14.0 **STEEL**

14.1 **Type of Steel**

14.2 **Reinforcement steel for all locations and structures except sump :**

Steel bars for concrete reinforcement wherever shown on drawings shall be TMT bars produced by Thermo Mechanical Treatment Process (Grade Fe 500 or Fe 500D), meeting all requirement of IS-1786-2008. Wherever mild steel reinforcement bars are shown on drawings these shall be amended to read as TMT bars of same dia or nearest higher dia and accordingly to be provided.

14.3 **Structural Steel -** Structural steel shall be any of the following two types depending upon the use :-

14.4 **Structural Steel E 250 (Fe 410W A & B)** :

All types of structures including those subjected to dynamic loading and where fatigue, wide fluctuation of stresses, reversal of stresses and great restrain are involved e.g. residential buildings, office building, workshop building, platform roofs, foot over bridges, transmission towers, crane gantry girders, road bridges etc.

14.5 **Structural Steel E 165 (Fe 290)** :

Grills / guard bars, hold fasts, railing, fencing, steel windows and steel doors etc.

PARTICULAR SPECIFICATIONS SECTION- II [CONTD/--]

14.6 **SOURCES OF PROCUREMENT**

14.7 **Structural Steel** : The structural steel E 250 (Fe 410W A & B) conforming to IS-2062 shall be procured from the following manufacturers :-

Sl No	Company Name	Brand	Type of Steel
1)	Rashtriya Ispat Nigam Limited	RINL	Structural Steel (Angle, Bream, Column, Channel. Plate)
2)	Tata Iron & Steel Company (TISCO or TATA STEEL)	TATA	-do-
3)	Steel Authority of India Limited	SAIL	-do-
4)	M/S Jindal Steels and power Ltd	JINDAL	-do-

14.8 Structural Steel E 165 (Fe 290) conforming to IS:2062 can be procured by the contractor from main producers / approved secondary producers / BIS marked manufacturer's or their authorized dealers at the option of the contractor.

14.9 **Reinforcement Steel** : TMT Steel Bars shall be procured from the following manufacturers :-

Sl No	Company Name	Brand	Type of Steel	Valid Upto
1)	Rashtriya Ispat Nigam Limited	RINL	-All-	-
2)	Tata Iron & Steel Company (TISCO or TATA STEEL)	TATA	-All-	-
3)	Steel Authority of India Limited	SAIL	-All-	-
4)	M/S Jai Balaji Industries Ltd	BALAJI SHAKTI	Fe 500, Fe 500D	Feb 2023
5)	M/S Shyam Steel Industries	SHYAM	Fe 500, Fe 500D, CRS	Jan 2023
6)	M/S Steel Exchange India Ltd	SIMHADRI TMT	Fe 500, Fe 500D, HSCRM	Jan 2023
7)	M/S Jindal Steels and Power Ltd	JINDAL PANTHER	Fe 500, Fe 500D, Fe 550, Fe 550D & CRS (8-40mm)	Feb 2023
8)	M/S SRMB Srijan Pvt Ltd	SRMB	Fe 500, Fe 500D, Fe 550, Fe 550D (8-32 mm dia)	Aug 2022
9)	M/S Shri Bajrang Power & Ispat Ltd	GOEL TMT	Fe 500, Fe 500D (8-32 mm dia)	07 Jan 2020
10)	M/S JSW Steel Ltd	NEOSTEEL	Fe 500, Fe 500D, Fe 550D, CRS (8-40 mm dia)	06 Feb 2021
11)	M/S Shyam Metalics & Energy Ltd	SEL	Fe 500, Fe 500D (8-32 mm dia)	Dec 2020
12)	M/S Kamachi Industries Ltd	KAMACHI	Fe 500, Fe 500D, Fe 550, Fe 550D, HCRM (8-40 mm dia)	May 2021
13)	M/S Real Ispat & Power Ltd	G K TMT	Fe 500, Fe 500D (8-36 mm dia)	Jan 2020
14)	M/S Super Smelters Ltd, Kolkata	SUPER SHAKTI	Fe 500, Fe 500D, Fe 550 (8-32 mm dia)	Feb 2020
15)	M/S Electrotherm (India) Ltd	ET TMT	Fe 500, Fe 500D & CRS (8-32 mm dia)	Jan 2021
16)	M/S Adhunik Industries Ltd	ADHUNIK Fe 500 SD	Fe 500D (8-32 mm dia)	04 Mar 2021
17)	M/S Gallantt Metal Ltd	GALLANT TMX	Fe 500, Fe 500D, CRS (8-32 mm dia)	05 Jul 2022
18)	M/S Rashmi Metaliks Ltd	RASHMI TMT	Fe 500 (8-32 mm dia), Fe 500D, Fe 550D (8-25 mm dia)	23 Aug 2022

However, reinforcement steel, Structural steel and Galvanised Steel sheets and Fabric reinforcement for concrete may be procured from authorized dealers of other main producers in case of value not exceeding Rs.7.5 lakhs.

14.10 **Galvanized steel sheets & Fabric reinforcement for concrete** :

(a) These shall be procured from main producers i.e SAIL, Rashtriya Ispat Nigam Ltd, TISCO or from BIS marked manufactures at the option of the contractor.

(b) The documents in support of purchase of steel/BIS Licence shall be verified by Engr-in-Charge & GE.

14.11 **TESTING OF REINFORCEMENT STEEL BY MANUFACTURER** :

The manufacturer shall carry out inspection & testing of steel in accordance with the relevant BIS provisions. The contractor shall submit the producer's (not from his authorized conversion agent) test certificate & Purchase Vouchers in original along with the test sheet giving the result of each mechanical test as applicable & the chemical & metallurgical composition of the steel or authenticated copy thereof duly signed by manufacturer with each consignments as per Appendix **'C'.**

14.12 **DOCUMENTATION** :

14.12.1 The contractor shall submit original purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE along with the relevant documents before acceptance. The original purchase vouchers and the test certificates shall be defaced by the Engineer-in-Charge and kept on record in the office of the Garrison Engineer (GE) duly authenticated and with cross reference to the consignment number recorded in the Steel Acceptance Register. The Steel Acceptance Register shall be signed by the JE (Civil), Engineer-in-Charge, GE and the contractor. The contractor shall maintain schedule of supply of steel for each consignment.

14.12.2 The Engineer-in-Charge shall record details of steel procurement and testing in "Steel Acceptance Register" and in the format as given at Appendix **'C'**, after due verification and send a certified true copy of test sheet to GE for his records.

14.12.3 GE / CWE shall organize independent testing or random samples of steel drawn from various lots from any of the organizations as listed for testing of cement as per the recommended minimum frequency. The elongation shall not be less than 18%. The cost of samples and transportation shall be borne by the contractor. The cost of testing shall be borne in accordance with Condition 10A of IAFW-2249 i.e. testing charges shall be borne by the Deptt if the test results are found to be in order, otherwise these shall be borne by the Contractor. Records of such checks would be maintained in the steel test register.

14.12.4 Any particular bar / steel section of any consignment not meeting the requirement shall be rejected and the same shall be removed from the site by the contractor at his own cost and the contractor shall have no claim on this account.

14.13 **TESTING OF REINFORCEMENT STEEL & STRUCTURAL STEEL E 250 (FE 410W A & B)**

14.13.1 **High Strength deformed Steel / TMT bars** :

Nominal mass test, tensile test, bend test and rebend test shall be carried out in National Test House, Govt. approved lab or any Govt Engineering colleges.

14.13.2 **Mild Steel Bars** :

Nominal Mass test, Tensile test and bend test shall be carried out as per the minimum frequency mentioned in Para 12.6.3 here-in-after all as per IS-432 (Part-I) – 1982.

14.13.3 **FREQUENCY FOR NORMAL MASS, TENSILE AND BEND TESTS**

A. **REINFORCEMENT STEEL**

NOMINAL SIZE OF BARS	**QUANTITY**
(i) Under 10 mm	1 sample consisting of 3 specimens for each test from each size of bars for every 25 ton or part thereof.

(ii)	10 mm to 16 mm	1 sample consisting of 3 specimens for each test (inclusive) from each size of bars for every 35 ton or part thereof.
(iii)	Over 16 mm	1 sample consisting of 3 specimens for each test from each size of bars for every 45 ton or part thereof.

B. **STRUCTURAL STEEL E 250 (FE 410W A & B)**

(i)	Tensile Test	One test for every 25 tonnes of steel or part thereof
(ii)	Bend Test	One test for every 10 tonnes of steel or part thereof

If any test result of any particular size of bar / section of steel of any consignment is not found satisfactory as specified in relevant IS, tests may again be carried out (if desired by contractor) for that particular size of bar / section of steel. Even after carrying out tests again if that size of bar / section of steel is not found satisfactory, the contractor shall remove the same at his own cost and no claims of contractor shall be entertained on this account. The testing charges for testing again shall be borne by the contractor.

14.14 **STORAGE, ACCOUNTING, PRESERVATION AND MAINTENANCE OF STEEL** :

Refer Clause 4.13 of IS-4082-1996 in this respect. The storage, accounting, preservation and maintenance of steel supplied by the contractor shall be done as per standard engineering practice till same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate / amount quoted by the tenderer. The GE shall inspect at regular intervals to verify the steel being brought at site is stored, accounted, preserved and maintained as per the norms. The steel shall be stored so as to differentiate each consignment separately. If GE is not satisfied with the storage / preservation of any size of bar / section of steel, he may order for any test(s) of steel as applicable for that size of bar / section of steel and as specified in tender documents / relevant Indian Standard to re-check the acceptability criteria for the same. The contractor shall bear the cost of necessary testing in this regard and no claim whatsoever shall be entertained.

14.15 **MEASUREMENT AND PAYMENT OF STEEL** :

14.15.1 The entire quantity of all steel items shall also be suitably recorded in the measurement book for "record purposes" as "not to be abstracted" before incorporation in the work and shall be signed by the Engineer-in-Charge and the Contractor. The nominal mass conversion factors for various steel section / size of finished bars as given in relevant IS codes shall be considered standard for measurement. The contractor shall plan to procure steel considering the time required for approval and testing well in advance. No extension of time on grounds of delay in procurement or testing / retesting of steel shall be admissible.

14.15.2 The payment of steel shall only be allowed after production of original purchase vouchers and test certificates from the main producer / secondary producer by the contractor for each consignment of steel and results of testing carried out by the Department (if any) are found satisfactory.

14.16 **ASSEMBLY AND PLACING OF REINFORCEMENT BARS** :

14.16.1 Reinforcement bars shall be TMT bars and shall be provided as indicated in drawings. For the purpose of pricing deviation involving reinforcement bars, the length of each bar shall be considered as 10 (Ten) meters for calculating laps unless otherwise shown on drawings.

14.16.2 Assembly and placing of reinforcement bars shall be as specified in Clause 12.1, 12.2, 12.3, 12.5 and 12.6 of IS-456-2000 and as shown in drawings. In case of any variation the drawings shall be followed. Cover to reinforcement bars shall be as per Clause 26.4.2.1, 26.4.2.2 and Table – 16 (for moderate exposure) of IS 456-2000.

14.16.3 In case of pricing deviations 10 mm and over dia TMT bars as stirrups shall be priced at the rate of TMT bar stirrups of 5 mm to 10 mm in MES Schedule Part – II subject to Contractor's percentage.

14.17 **MILD STEEL WIRE**

14.17.1 Mild steel wire shall be all as specified in Schedule 'A' and as per clause No. 10.31 on Serial Page No. 214 of SSR Part-I 2009.

14.18 **EXPANDED METAL STEELS**

14.18.1 Expanded metal steel shall be all as specified in Schedule 'A' and as per clause No. 10.35 on Serial Page No. 215 of SSR Part-I 2009.

14.19 **MILD STEEL GUARD BARS/OTHER STEEL ITEMS**

14.19.1 Mild steel guard bars shall be 10 mm square bars of Fe-310-0 or Fe-410-0 quality. Guard bars shall be provided to all windows & vents ass shown on drawings. The guard bars including windows shall be painted with 2 coats of synthetic enamel paints over a coat of red oxide chrome primer.

14.19.2 Other steel items like steel doors with PGI sheet covering, railing in balconies and stair cases, water meter and switch board boxes etc shall be fabricated to the dimensions as shown on drawings and as directed by Engr-in-Charge. Welding, bolting, drilling holes etc shall be as per specified in MES schedule.

15. **FABRICATION & ERECTION OF STEEL TRUSS**

15.1 **GENERAL**

15.1.1 Refer to Clause No. 10.16 of MES Sch Part-I as applicable and as directed by the Engr-in-Charge.

15.2 **COMPONENT MEMBERS**

15.2.1 All Component Members shall all as specified here-in-before.

15.3 **FABRICATION**

15.3.1 Fabrication of all Components shall be carried out all as specified here-in-before.

15.4 **ERECTION OF TRUSSES**

15.4.1 Erection shall be carried out as specified in Clause No. 10.16 of MES Schedule Part-I.

15.4.2 The Trusses shall be lifted only at nodes. The Trusses shall not be aligned at the Apex. It shall be lifted by slinging at Two Mid Points of Rafter, which shall be temporarily braced by suitable Section. After Trusses are placed in position, Purlins and Wind Bracing shall be fixed as soon as possible.

15.4.3 Contractor's particular attention is drawn to IS-7205 (Safety Code for erection on Structural Steel Work). Anchor bolts for fastening of steel structure shall be set in designed position and grouted alongwith foundation. Also refer Clause No. 10.16.2 and 10.16.3 of MES Sch Part-I.

15.4.4 The erected parts of structure shall be stable during all the stage of erection and the structural elements to be erected, shall be stable and strong to bear erection load.

15.4.5 Working on the already erected structure is permitted only after they are finally fixed. Erection of structures of each tier of high structure shall be executed only after the relevant fastening of lower ties by the permanent or temporary fastening devices as clarified for safety.

16.0 **FLOORING**

16.1 **GENERAL**

16.1.1 Refer Clause No. 13.2, 13.4, 13.5, 13.8, 13.14, 13.21, 13.23, 13.24, 13.25, 13.27, 13.32, 13.38, 13.39, 13.40, 13.46 and 13.47 of MES Schedule Part-I are to be adopted for Materials and Workmanship of laying Floor/ Skirting/Dado.

16.1.2 Floors shall be laid to levels or to falls as shown on drawings and as directed by Engineer-in-Charge. Floor finish shall be extended over dwarf wall, doors and other opening.

16.1.3 Steps and connected ramps shall have the same type of floor as adjoining floors of building unless otherwise specifically mentioned and shall have chequered finish in addition.

16.1.4 Where two different types of floors finish are provided on either side of the opening the superior
floor finish shall be extended over the opening.

16.1.5 Finishing layer of floor shall be carried out through all opening and over dwarf wall.

16.1.6 PCC floor shall be finished even and smooth using extra cement unless otherwise specifically mentioned in drawings of schedule of finishes or here-in-after.

16.1.7 Polythene film shall conform to IS-2508, 150 micron thick of 01.0 Grade. If full width of polythene film is not available, the same shall be laid with an overlap of 150mm pasted with the resin solution.

16.1.8 Sub floor may not be laid in panels. Under layer and toping layer of casting PCC/terrazzo cast-in-situ floor shall be laid in square or rectangular panel as directed by Engineer-in-Charge. Length of panel shall not exceed 2.0 Metre. Length width ratio shall not exceed 1.5 times in rectangular panels. However, area of each panel shall not exceed 2.0 Square Metre as specified. Cement concrete in sub base or sub floor and top wearing coat shall be laid separately and not monolithically.

16.2 Cement: cement shall be as specified for concrete work.

16.3 Aggregate: Stone aggregate and sand shall be as specified for cement concrete.

16.4 **PCC FLOORS**

16.4.1 **GENERAL**: Refer paras 13.25, 13.27 and 13.32 of MES Schedule Part-I.

16.4.2 PCC floors unless indicated, shall be considered as 40mm thick PCC of grade M-15 (Nominal Mix), laid in panels with dividing strips and shall be finished even and smooth using extra cement laid over 75mm thick PCC 1:4:8 type D-2 over 200 micron polythene film over 75 mm thick sand cushioning over rammed earth.

16.4.3 **PCC SKIRTING**: Unless otherwise indicated on drawings height of skirting shall be 10cm. Cement plaster skirting shall be provided wherever shown on drawings. It shall be 15 mm thick in CM (1:3) finished even and smooth with using extra cement, tapered on 45/60 degree with plastered surface.

16.4.4 All PCC floors shall be finished even and smooth using extra cement/without using extra cement as mentioned in Schedule 'A' before placing the concrete mix as floor toping coat cement slurry @ 3 Kg/Sq. M shall be thoroughly brushed into the prepared, surfaces of base concrete just ahead of the finish. The coat for the same shall be deemed to be included in the rate for floor topping.

16.11 **PCC PLINTH PROTECTION**

16.11.1 Unless otherwise directed, PCC plinth protection shall be provided with 50mm thick PCC (1:3:6) type C-1 over 75 mm thick (consolidated thickness) broken stone or boulder hard core of broken stone over rammed earth. The surface of plinth protection shall be finished even and smooth without using extra cement. The concrete in plinth protection shall be laid in alternate bays of length not exceeding 3m and shall be laid in slope 1 in 12. The joints shall be 12 mm thick filled with bitumen mastic.

17.0 **PLASTERING AND POINTING**

17.1 Cement: Cement shall be as specified for concrete work.

17.2 Sand: Refer paras 14.5, 14.5.1, and 14.5.2 of MES Schedule Part-I.

17.3 Water: Water shall be as specified for concrete work.

17.4. The thickness of plaster mentioned in Sch 'A' is finished thickness exclusive of dubbing out, plastering and dubbing out shall however, be done in one operation.

17.5 **WPC:-** Integral water proofing compound (WPC) shall conform to IS -2645 and the brand shall be got approved in writing from the GE. Preparation of mix & method of mixing shall be as per manufacturer's instructions. While pricing deviation for omission, qty shall be considered as 3% by weight of cement.

18.0 **PAINTING**

18.1 Paints and allied materials: Refer Para 17.2 of MES Schedule Part-I.

18.2 Painting work Generally: Refer para 17.3 of MES Schedule Part-I.

18.3 Painting woodwork: Refer para 17.6 of MES Schedule Part-I. Filler coat is not required to be applied. Where painting is indicated in BOQ, it shall be done in 3 coats viz. primer, under coat & finishing coat.

18.4 Paints shall be of Ist grade quality of the approved makes and shall be procured by the contractor in sealed containers.

18.5 Primer for different surfaces i.e. wooden / steel / plastered /concrete surfaces shall be of the appropriate type and shall be of same make as the paint to be applied over respective surfaces

18.6 Primer for different surfaces i.e. wooden / steel / plastered /concrete surfaces shall be of the appropriate type and shall be of same make as the paint to be applied over respective surfaces

18.7 The contractor shall inform the Accepting Officer, within three weeks of the acceptance of the tender, the brand names of the manufacturers of paint proposed to be used in the works and submit sample thereof and obtain prior written approval of the Accepting Officer before their use in work. Paints for priming coat under coat and finishing coat shall be of the same manufacturer. Tint of paint will be approved by Accepting Officer.

18.8 The paints shall be procured by contractor direct from the manufacturers of their authorized dealers and original paid vouchers bearing machine number for the full quantity showing the name of manufacturer and brand shall be produced to Engineer-in-Charge.

18.9 Preparation of old painted surfaces shall be done in accordance to para 17.10 of SSR Part I 2009. Application of synthetic enamel paint to wooden and steel surfaces shall be done in accordance to the relevant items of Sch 'A' and as specified in Paras 17.6 & 17.8 of SSR Part I, 2009 respectively, under close supervision of the Engineer-in-Charge.

18.10 French polishing shall be done in accordance to para 17.17.4 of SSR Part I 2009.

18.11 Varnishing shall be done in accordance to para 17.17.3 to 17.7.3.2 of SSR Part I 2009.

19.0 **PLUMBING**

19.1 **MILD STEEL GALVANISED TUBES (PIPES) AND FITTINGS**:- Refer para 18.4 of MES Sch Part – I. The pipe & fittings shall be ISI marked & shall be of incorporated in work duly approved by GE.

19.2 **BIB TAPS, PILLAR TAPS, STOP VALVE AND GUN METAL GATE VALVE**: - Refer para 18.4 of MES Sch Part – I. Screw down bib taps & stop cocks shall be of cast copper alloy body bright finished, ISI marked.

19.3 **WORKMANSHIP**

19.3.1 **GENERAL REQUIREMENT**: - Refer paras 18.41 of MES Schedule Part – I. Generally layout of water supply work is shown on drawings. The exact position of water supply and fittings shall be as directed by the Engr-in-Charge. The mounting height of various fittings shall be as directed by Engr-in-Charge.

19.3.2 **LAYING OF GI PIPE**:- Refer para 18.46 of MES Schedule Part – I. Union shall be provided at suitable places as directed by the Engr-in-Charge so that in case of repairs etc. long length of pipes are not disturbed.

19.3.2.1 Uncovered threaded portion of pipe shall be painted with approved paint. GI tubing in chases shall be painted with two coats of anti corrosive bituminous paint.

19.3.2.2 The contractor shall use proper bends/elbows/tees at turning/corners. Bending of pipe is not permitted.

19.3.2.3 The contractor shall provide screwed plug to all open bends of fittings at the end or days work and at time when the work is stopped.

19.3.3 **FIXING OF WATER SUPPLY FITTINGS**:- Refer paras 18.62, 18.63, 18.64 and 18.66 of MES Schedule Part – I. Fixing shall be done as specified in SSR Part – 2009 and as directed by Engineer-in-Charge.

19.4 **TESTING**:- Refer clause 18.55 of MES Schedule Part – I. The water supply installation shall be tested hydraulically at a pressure of 2.5 Kg/sqcm in the presence of Engineer-in-Charge for at least 24 hours continuously to ensure leak proof connections. All arrangements for testing shall be made by the contractor at his own cost. Any leakage/defects noticed shall be rectified to entire satisfaction of Engineer-in-Charge

20.0 **INTERNAL ELECTRIFICATION**

20.1 **General**: Refer paras 19.2 of Schedule Part-I MES. All work of internal electrification shall be executed by fully qualified licensed electricians all as per the latest Engineering practice and latest IS codes/IS Rules.

20.2 **WORK IN ACCORDANCE WITH SPECIFICATION:-** Work under this contract shall be carried out as per brief description given in Schedule 'A', Particular Specifications, **NATIONAL ELECTRICAL CODE 2011 (NEC-2011)** and as shown on drawings. Work shall also be in conformity with **NATIONAL ELECTRICAL CODE 2011** and MES Schedule which are forming part of the contract. The entire work shall be carried out properly with due care and under the supervision of competent technical personnel.

20.3 The materials shall be as follows: -

Ser No	Materials	Reference to paras of MES Schedule Part-I.
(a)	Cables, Cords and earthling leads.	19.24
(b)	Cables for internal wiring light power and sub main.	19.25
(c)	Conduit and conduit fittings.	19.29
(d)	Plug Gutties	19.30
(e)	Screws and fastening link clips	19.31
(f)	Ceiling Rose	19.32
(g)	Socket outlet	19.40
(h)	Lamp holder	19.41
(j)	Distribution board	19.44
(k)	Switch fuses	19.45
(l)	MCB Distribution Boards	19.46.1.1,19.46.1.2 & 19.46.1.3
(m)	MCBs	19.46.2
(n)	MCBs Isolators	19.46.3
(o)	Plug and Socket DBs	19.46.4

20.4 PVC casing and capping wiring shall be as per IS: 14927 (Part-I&II)-2001, of approved make and fire retarded quality. Wall thickness shall be 1.20 mm (Minimum). The channel space shall be adequate to accommodate the cable laid in it. The screws shall be cadmium-coated iron-screws rigidly fixed to gutties or with Detofix at least 15 cm apart. The corner bridges shall be made neatly angle shaped.

20.5 Distribution board, main switches: Refer Para 19.107 of MES Schedule 'A' Part-I.

20.6 Accessories switches, lamp holder, ceiling roses, socket outlets, Refer paras 19.114 of MES Sch Part-I.

20.7 **NON METALLIC (PVC) CONDUIT AND CONDUIT ACCESSORIES:-**

20.7.1 Refer clause 19.29 of MES Schedule Part-I, 2009. Conduit and conduit accessories shall conform to relevant IS i.e., IS-9537 (Part-III) 1983 and IS-3419-1976. Conduit shall be non-metallic PVC conduit and shall show no appreciable un-evenness. Interior and ends of conduit and conduit accessories shall be free from bushes, ducts fins and the like which may cause damage to the cables. Each length of PVC conduit shall be marked with the following :-

(a) Manufacturer's name or trade mark. (b) Country of manufacture
(c) Nominal size of the conduit.
(d) Each length of conduit shall also be marked with the ISI certificate mark.

20.7.2 PVC flexible conduit of size as directed by the Engineer-in-charge shall be provided as required without any extra cost.

20.8 **PVC INSULATED 1100 VOLTS GRADE CABLE (PVC SHEATHED/UNSHEATHED)**

20.8.1 The conductor shall be of stranded copper wires. The insulation shall consist of compounded polyvinyl chloride. The thickness of insulation shall not be less than nominal value mentioned in appropriate table of IS-694-1964. The composition of the compound of sheath shall be polyvinyl chloride. The compound shall be designed to offer adequate mechanical protection. The value of thickness of sheath shall be in accordance with those specified in appropriate tables. The insulation on & sheath shall be applied by the extrusion process and each shall form a compact homogeneous body.

20.9 **SWITCH BOXES**

20.9.1 Switch boxes suitable to house switches of required ratings, and fan regulators as required shall be provided. In case the number of switches in one box is not tallying with that available in standard manufacturer, the box accommodating the next higher of switches shall be provided without any extra cost. In case fan regulator/regulators is/are to be provided at a later dated, suitable provision for accommodating such regulators shall be made in the switch boxes and blank of covers shall be provided without any extra cost.

20.9.2 Switch boxes shall be so designed that accessories are mounted on a grid plate with tapped holes for brass machine screws leaving ample space at the back and on the side for accommodating conductors, check-nuts and screwed bushes at conduit entries etc.. The grid plates and MS boxes shall be fitted with a brass earth terminal. Boxes shall be attached to conduits by means of check-nuts on either of their walls. Moulded front covers made from high impact resistant and ultra violet stabilized engineering plastics shall be fixed by means of counter sunk chromium plated brass machine screw. No timber shall be used for any supports. Switch boxes shall be located with bottom at 1200mm above floor level unless otherwise indicated.

20.10 **BOXES FOR SOCKETS**

20.10.1 Outlet boxes shall be suitable for housing switches socket outlets and any other outlet as required. These shall be so designed that accessories are mounted on a grid plate with tapped holes for brass machine screws leaving ample space at the back and on the sides for accommodating conductors, check nuts and screwed bushes at conduit entries etc.. The grid plates and MS boxes shall be fitted with a brass earth terminal. These shall be attached to conduits by means of check nuts on either sides of their walls. Moulded front covers made from high impact resistant, flame retardant and ultra violet stabilized engineering plastics shall be used to mount the outlets and shall be fixed to the outlet MS boxes by means of counter sunk chromium plated brass machine screws. No timber supports shall be used. Boxes shall be located at skirting level or bottom at 1200mm from floor or inside raceways on laboratory work table., as indicated in drawings and / or as directed.

20.11 **SWITCHES/SWITCH SOCKET OUTLET/SOCKET OUTLET**

20.11.1 The base shall be made of vitrified ceramic material or tough non ignitable molded insulating material. The covers, cover plates and actuate members shall be of tough, non-ignitable insulating material. Current carrying parts shall be made of brass, copper phosphor bronze, aluminium alloy or any other suitable material, springs shall be of corrosion resistant metal. Attachment fitting screw and other non-current carrying Paris shall be of mild steel aluminium alloy or insulating material. All socket outlets and switch socket combination shall have ISI certification mark.

20.12 **LIST OF BIS CERTIFIED PRODUCTS TO BE INCORPORATED IN WORKS**

(a)	Ceiling Rose	IS-371
(b)	Socket outlet 3 pin plug and socket	IS-1293
(c)	Rigid non-metallic conduit	IS-9537
(d)	Copper stranded conductor	IS-694
(e)	Switch fuse (Main switch)	IS-4064
(f)	Single core cable polythene Insulated and PVC sheathed	IS-1596
(g)	Distribution Fuse Boards and Cutouts	IS-2675
(h)	Rigid non-metallic casing & capping	IS-14927

21. **ENERGY EFFICIENT LED LIGHT FITTINGS** :- The contractor shall furnish and install the complete LED luminaries as described in BOQ as as described here-in-after complete with all accessories, LED lamps with suitable current control driver circuit including mounting arrangement for surface mounted type, recessed type and ceiling mounting arrangements.

21.1 **STANDARDS** :- The LED light fixtures/luminaires & their associated accessories such as lamps/ tubes, reflector, housings, energy efficient ballasts/drives etc shall comply with the latest applicable standards as under:-

IS:16101	:	General Lighting-LEDs and LED modules – Terms and Definitions
IS:16102-1	:	Self-Ballasted LED Lamps for General Lighting Services Part I safety requirements
IS:16102-2	:	Ballast type LED lamp performance requirement, lamp and accessories.
IS:16107-2-1	:	Luminaires performance LED luminaires.
IS:16103-2	:	LED modules performance requirement electric lamps and accessories.
IEC 62031	:	LED modules for general lighting – safety requirements.
EN 61547	:	Equipment for general lighting purpose – Electro Magnetic Compatibility (EMC) immunity requirement.
IEC 60598-2-1	:	Fixed general purpose luminaires.
IEC 60598-1	:	Luminaires – General requirement and tests.
IEC 61000-3-2	:	Electro Magnetic Compatibility (EMC) – Limits for Harmonic current emission – (equipment input current – less than or equal to 16 Amps per phase.
IEC 61347-2-13	:	Lamp control gear – particular requirement or DC and AC supplied electronic control gear for LED modules.
IS:10322	:	Specification or the Electro luminaires.
IS:4905	:	Method for random sampling.
LM 79	:	LED luminaire photometry measurement.
LM 80	:	Lumen maintenance.
IEC 62384	:	DC or AC supplied electronic control gear for LED modules – performance requirement.
IEC / PAS 62612	:	Self-ballasted LED lamps for general lighting service – performance requirement.
IEC 60529	:	Classification of degree of protections provided by enclosures

21.2 **GENERAL REQUIREMENT** :-

(a) The LED light fixtures / luminaires shall be designed for continuous trouble-free operation under hot humid atmospheric conditions, at the specified ambient temperature, without reduction in lamp life or without deterioration of materials and internal wiring.

(b) The LED light fixtures / luminaires shall be designed so as to facilitate easy maintenance, including cleaning, replacement of lamps / drives, starters etc.

(c) Connections between different components shall be made in such a way that they will not work loose by small vibration.

(d) LED light fixtures / luminaires shall be supplied complete with lamps.

(e) LED light fixtures / luminaires and accessories shall be designed to have low temperature rise. The temperature rise above the ambient temperature shall be as indicated in the relevant standards.

21.3 **EARTHING** :-

(a) Each light fixture / luminaire shall be provide with an earthing terminal suitable for connection to the earthing conductor.

(b) All metal or metal enclosed parts of the housing shall be bonded to the earthing terminal so as to ensure satisfactory earth continuity throughout the fixture.

PARTICULAR SPECIFICATIONS SECTION- II [CONTD/--]

21.4 **PAINTING / FINISH :-**

(a) All surfaces of the fittings shall be thoroughly cleaned and degreased.

(b) The fittings shall be free from scare, rust, sharp, edges and burrs.

(c) The housing shall be store-enamelled / epoxy stove-enamelled / vitreous enameled or anodised as indicated under various types of fitting.

21.5 **ACCESSORIES FOR LIGHT FIXTURES** :-

21.5.1 **LED LAMPS** :-

21.5.2 **Technical specification of LED Lighting Fixtures (Internal Fittings)** :-

No	Parameters	Value
(A)	**Site Parameters**	
1	Ambient Temperature	**0°C to 50°C**
(B)	**Electricity Efficiency Management / Electronic Driver**	
1	Input Operating Voltage	180 to 270 Volts, AC 50H_z±3
2	AC Power Factor	Not less than 0.95
3	Efficiency of driver	More than 85%
4	THD (AC current 180 V to 270 V)	Less than 10%
5	LED Drive Current	Not more than 300mA / W
6	LED Efficacy	As mentioned in BOQ
(C)	**Optical Management**	
1	LED Make	**Osram/Cree/Nichia/Phillips Lumileds**
2	Colour Temperature	As specified in individual schedule of items
3	LED life	Lumen maintenance of the lamp shall not be less than 80% of the initial lumen after 20000 burning hours and 70% of the initial lumen after 50,000 burning hours. The initial lumen will be taken after 100 hours aging.
4	E Avg, 9 point method	As per BIS Standard.
5	CRI	More than 80 or as specified in individual schedule of items whichever is better.
6	Illumination Uniformity	As per BIS Standards. The illumination should be uniform without Dark Bands or abrupt variations and have proper glass for soothing to the eyes.
7	Average lighting angle	160° or more.
(D)	**Thermal Management**	
1	Frame of fixture, Heat Sink	Wherever applicable Single Piece die cast High grade Aluminium grade >6000.
2	Jn. Temperature of LED at 25°C	≤65°C
3	Heat Sink temperature rise above ambient	≤30°C
(E)	**General parameters**	
1	Surge Protection	Yes to be provided separately or to be in built not less than 3 kv or as specified in individual schedule of items whichever is suitable.
2	Cover	Non-inflammable toughened glass 4mm thick with silicon gaskets
3	Overall system efficiency	Not less than 95%
4	IP Clause & Environment Protection	IP 20 and above.
5	Starting time	Instant on
6	Fire protection	Connecting wires used inside the luminaries Smoke Halogen Free, fire retardant e-beam cable and fuse protection shall be provide in input side. All the material used in the luminaries shall be halogen free and fire retardant confirming to UL94.

21.5.3 **TEST REPORTS** :-

(a) The contractor shall submit the LM-80 test report of the light fittings, conforming to the applicable IEC/PAS 62717 for LED Modules Performance and LM-79 conforming to the applicable IEC/PAS 62722-2-1 for LED Luminaries Performance issued by NABL Accredited/Govt Approved Testing Laboratories/in house testing laboratories, issued on or after 01st Apr 2012 alongwith the copy of the letter issued by NABL/Govt authority for accreditation.

(b) **REFLECTORS** :-

(i) The reflectors shall be made of CRCA sheet steel / aluminium / silvered glass as indicated for above mentioned fittings.

(ii) The thickness of steel / aluminium shall comply with relevant standards. Reflectors made of steel shall have stove enameled / vitreous enameled / epoxy coating finish.

(iii) Aluminium used for reflectors shall be anodized / epoxy stove enameled / mirror polished.

(iv) Reflectors shall be free from scratches or blisters and shall have a smooth and glossy surface having an optimum light reflecting co-efficient such as to ensure the overall light output specified by the Manufacturer.

(v) Reflectors shall be easily removable from the housing for cleaning and maintenance without disturbing the lamps and without the use of tools. They shall be securely fixed to the housing by means of positive fastening device of captive type.

(c) **BALLASTS / DRIVES** :-

(i) The ballasts/drivers shall be designed, manufactured and supplied in accordance with the relevant standards. The ballast/drivers shall be designed to have a long service life and low power loss.

(ii) Ballasts/drivers shall be mounted using self locking, anti-vibration fixings and shall be easy to remove without de-mounting the fittings. They shall be in dust-tight, non combustible enclosures.

(iii) The Ballasts/driver shall be low loss electronic type and suitable to operate at 180V – 270V. End connections shall be brought out in a suitable terminal block, rigidly fixed to the ballast enclosure.

(iv) Separate ballast/driver for each lamp shall be provided in case of Multi Lamp Fittings.

(d) **LAMP HOLDERS** :-

(i) Lamp holders, if any shall comply with relevant standards. They shall have low contact resistance, shall be resistant to wear and shall be suitable for operation at the specified temperature without deterioration in insulation value. They shall hold the lamps in position under normal condition of shock and vibration met with in normal installation and use.

(ii) Lamp holders for the LED tubular lamps shall be of the spring loaded bi-pin rotor type, Live parts of the lamp holder shall not be exposed during insertion or removal of the lamp or after lamp has been taken out. The lamp holder contact shall provide adequate pressure on the lamp cap pins when in working position.

(iii) The lamps shall be capable of withstanding small vibrations and the connections at lead in wires shall not break under such circumstances.

(iv) Lamps shall conform to relevant standards and shall be suitable for supply voltage and frequency specified.

(e) **CAPACITORS** :-

(i) The capacitors shall have a constant value of capacitance and shall be connected across the supply of individual lamp circuits.

(ii) The capacitors shall be suitable for operation at supply voltage as specified and shall have a value of capacitance so as to correct the power factor of its corresponding lamp circuit to the extent of 0.95 lag.

(iii) The capacitors shall be hermetically sealed preferably in a metal enclosure to prevent seepage of impregnate and ingress of moisture.

21.6 **WARRANTY**

21.6.1 LED light fixtures/luminaires supplied shall be warranted for a period of 36 months from the date of commissioning or 39 months from the date of material in certified good condition whichever is earlier. The contractor shall ensure to provide the minimum essential services to GE during the warranty period.

21.6.2 The contractor shall repair/replace the defective pieces within 7 days of the receipt of complaint. The contractor shall repair/replace the defective material at his own expenses to the satisfaction of GE/Engr-in-Charge during the warranty period. Warranty cards for all fixtures shall be duly submitted alongwith the delivery of materials.

22.0 **EARTHING**

22.1 Earthing shall be done in accordance with latest IS: 3043 -1987 (Reaffirmed 2001), Code of practice for earthing and MES Schedule Part-I and IE rules. The contractor shall obtain approval of Engineer-in-Charge before commencement of earthing work and it shall be executed in the presence of an MES representative.

22.2 Normal earthing unless otherwise specified in Schedule 'A' shall be done with galvanized iron or steel plate 600 x 600 x 6mm as per Electrical Plate No 3 of MES Schedule Part – I. Depth of earth plate given in Schedule 'A' is the minimum which may be increased by the Engineer-in-charge depending the soil conditions. GI tubing used for earth lead protection & watering shall be of medium grade & light grade respectively.

22.3 LT switch gears/equipments, wherever to be earthed with normal earthing, shall be earthed with 32 x 6mm GI strip connection to the earth electrode. Two separate earth points shall be made in such cases for two independent paths to earth. In other cases each connection to electrode shall be of 8 gauge (4mm) GI wire as specified in Schedule 'A'.

22.4 The earth electrode shall be located at least 2 metres from the footing of nearest foundation, exact distance however being decided by GE at site considering the depth of excavation and structural safety. Distance between any two electrodes shall strictly be as per IS-3043. The continuity resistance of any electrode shall not exceed one ohm.

22.5 Test links (test point) shall be provided in earth circuit for carrying out resistance measuring test.

22.6 All joints in main earthing and between the main earthing strip and branch connections shall have the overlapping surfaces properly treated and jointed to prevent oxidation. Straight joints shall have overlap of 50mm. All joints shall be brazed. Form 6 mm dia copper rivets (2 in 2 rows) shall be used for jointing the main/branch earth strips.

22.7 Concrete chamber shall be PCC 13:6 type C-1. All internal surface of the chamber shall be given 15mm thick plaster in cement mortar 1:4. Funnel in chamber shall be made out of CI. It shall be leak proof and provided with wire gauge duly soldered.

22.8 Cover and frame for earth pit shall be as specified in Schedule 'A'.

22.9 Charcoal dust and salt filling shall be done in layers as shown in electrical plate No 3. Surplus soil shall be disposed off and site left clear and tidy on completion.

22.10 **GI/ALUMINIUM/COPPER STRIP**

22.10.1 GI strip shall be as mentioned in Sch 'A' and shall be incorporated in the work duly approved by GE.

23.0 **MILD STEEL GALVANISED TUBES (PIPES) AND FITTINGS**

23.1 Refer para 18.4 of MES Schedule Part – I. The pipe & fittings shall be ISI marked & shall be of incorporated in work duly approved by GE.

23.2 **WORKMANSHIP**

23.2.1 **GENERAL REQUIREMENT**: - Refer paras 18.41 of MES Schedule Part – I. Generally layout of water supply work is shown on drawings. The exact position of water supply and fittings shall be as directed by the Engineer-in-Charge. The mounting height of various fittings shall be as directed by Engr-in-Charge.

23.2.2 **LAYING OF GI PIPE**:- Refer para 18.50, 18.51 and 18.56 of MES Schedule Part-I. Unions shall be provided at suitable places as directed by the Engineer -in Charge so that in case of repairs etc, long length of pipe is not disturbed.

23.2.3 Uncovered threaded portion of pipe shall be painted with approved paint. GI tubing in chases shall be painted with two coats of anti corrosive bituminous paint.

23.2.4 The contractor shall use proper bends/elbows/tees at turning/corners. Bending of pipe is not permitted.

23.2.5 The contractor shall provide screwed plug to all open bends of fittings at the end or days work and at time when the work is stopped.

24.0 **DEMOLITION/DISMANTLING/TAKING DOWN**

24.1 Refer to relevant paras under section 21 of MES Schedule Part-I.

24.2 The contractor shall take proper precaution for demolition/dismantling work. If any damage caused due to dismantling/demolition to the existing structure, the contractor shall made good the same at his own cost without any extra cost to the Govt.

24.3 Unserviceable materials obtained from demolition/dismantling shall be property of the contractor and shall be credited as per schedule of credit. All demolished/dismantled materials shall be removed from the site of `the contractor as directed by the Engineer-in-Charge without any extra cost to the Govt.

25.0 **PURCHASE VOUCHER**

25.1 The contractor is advised to submit his original purchase voucher duly machine numbered in respect of the all materials as directed by E-in-C Branch latest .The materials shall be purchased from the authorized agents/dealers who are required to show on the cash memo /bill all the taxes etc and make and IS number as required. The purchase vouchers shall be produced to the Engineer in Charge for verification of materials equipment's before his permissions to incorporate the same in the work communicated to the contractor. Two certified copies of the same shall be made by the Engineer-in-Charge for reference and record.

26.0 **APPROVED NAMES OF MANUFACTURERS**

26.1 Approved names of manufacturers of various items/equipments to be incorporated in this work unless otherwise specified in Schedule 'A' and in particular specifications shall be as under and as specified here-in-before:-

(A) B/R ITEMS

Ser No	DESCRIPTION OF ITEM	MAKE
1	2	3
1	Synthetic Enamel Paint (1st Quality)	SHALIMAR PAINTS (SUPERLAC) / ASIAN PAINTS (APCOLITE) / BERGER (LUXOL) / J&N (BROLAC) / NEROLAC (IMPRESSIONS)
2	Precast Concrete Interlocking Paving Blocks / Precast Plain & Decorative/ Chequered Cement concrete Tiles etc	Anjali Tiles / Ultra Tiles / Terra Firma / Ecco Scope / Wonder Crete / Multiwyn / Bansal / A-One Tiles / Stylish Precast Pvt Ltd / Shyam Steel Industries Ltd
3	Factory made panel/ flush door and window	MP Wood Products /Doorking, Kolkata / Ambika Timber Works / Premier Wood Craft (P) Ltd / Jain Doors Pvt Ltd / M/S Modern Doors / M/S JOINERY WOOD PRODUCT / M/S Goel Brothers / M/S Surbhi Metal India Ltd /M/S Hindustan Housing Factory
4	SANITARY FITTINGS	CERA/HINDWARE/PARRYWARE/JAQUAR
5	CP BATHROOM FITTINGS	PRAYAG/JAQUAR/ESS ESS/PARRYWARE/HINDWARE

PARTICULAR SPECIFICATIONS SECTION- II [CONTD/--]

(B) WATER SUPPLY ITEMS

Ser No	DESCRIPTION OF ITEM	MAKE
1	2	3
1	GI Pipe	TATA / BANSAL / SURYA / PRAKASH / JINDAL / SURYA ROSHNI / SWASTIK / BST / ZENITH
2	GI fittings	ISI marked

(C) ELECTRICAL ITEMS

Ser No	DESCRIPTION OF ITEM	MAKE
1	2	3
1	PVC insulated 1100 volts grade cable for wiring (Copper)	FINOLEX / HAVELLS / POLYCAB / PLAZA / RR KABEL /HARSH / KEI / RICHA / MESCAB/ HPL / BCH / ANCHOR
2	PVC CASING CAPPING	PLAZA / PRESTO PLAST / FINOLEX / KALINGA / PRINCE / AKG / RICHA / HARSH / ANCHOR
3	PVC Conduits & Accessories	BEC / PRESTO PLAST / PLAZA / MODI / BERLIA / RICHA / KALINGA GOLD / AKG / PRINCE / INDO AMERICAN ELECTRICALS / HARSH / ANCHOR
4	Modular Switch, sockets and fittings	CRABTREE (THAMES) / ANCHOR (WOODS) / ABB / LEGRAND (MYRIUS) / North West / Honeywell MK Electric / PHILIPS
5	LED Luminaires	BAJAJ / PHILIPS / WIPRO / HAVELLS / CROMPTON / HALONIX / SURYA / C&S / HPL / JAQUAR / EVEREADY / PANASONIC / MAGIK / RASHMI / ORIENT ELECTRIC / SYSKA / SIMOCO TELECOMMUNICATIONS / SORIL INFRA RESOURCES
6	Steel conduits	BEC / KALINGA / AKG / PRAKASH / STEEL KRAFT / JINDAL
7	DB, MCB, Isolator	LEGRAND / C&S / ABB / L&T / SCHNEIDER/ HAGER/ HAVELLS / INDO ASIAN (OPTI PRO) / BCH / HPL
8	HT & LT Armoured/un-armoured XLPE insulated Cables	HAVELLS / POLYCAB / NICCO / CCI / UNIVERSAL/ KEI/ CRYSTAL / ASIAN (RPG) / GLOSTER/ RR KABEL / DYNAMIC

Note :-

1. The makes for a particular product specified anywhere else in the tender documents shall take precedence over makes given in this Specification.
2. The makes have been given trade wise only for the purpose of easily locating the items and otherwise makes given are applicable to the entire contract.
3. Wherever word "equivalent make" has been specified the option of equivalent make will only be considered after taking approval from Accepting Officer and only where products of brand/manufacturer specified are not available in market.
4. The make/ manufacturer of material(s) unless otherwise mentioned in this particular specifications/Schedule 'A' but required to complete the work shall be as per relevant IS of latest edition as approved by GE.

SIGNATURE OF CONTRACTOR **AGE (CONTRACTS)**
FOR ACCEPTING OFFICER

DATED :